Worked solutions to

Bayesian Statistical Methods

With Applications to Machine Learning — Reich & Ghosh (2nd ed.)

Chapter 2

From prior information to posterior inference

Every exercise is presented in the same order:

- the **exact** exercise text from the book;
- the problem **in plain words**;
- the **background** you need, recalled from scratch;
- the **key idea** that unlocks the problem;
- a **step-by-step solution** — every algebra step written out, with a *Why?* note whenever a step is not obvious;
- a **sanity check** of the answer;
- the boxed **answer**;
- **takeaways** and common mistakes.

All numbers and figures are reproduced by `ch02/code/ch02.R`.

Sai Sridhar · October 20, 2026

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Answers at a glance

1	$\frac{n\bar{Y}}{n+m} \pm 1.96\frac{\sigma}{\sqrt{n+m}}$; $m = 0$ gives exact 95% coverage for every μ.
2	Regular ≈ 0.200, World Series ≈ 0.371 HR/game; $\text{Prob}(\lambda_{WS} > \lambda_R) \approx 0.95$ — suggestive, not conclusive.
3	$\theta \sim \text{Gamma}(a, b) \Rightarrow \theta \mid Y \sim \text{Gamma}(a+1,\ b+Y)$.
4	$\lambda \mid \mathbf{Y} \sim \text{Gamma}(30.1, 6.1)$; $\text{Prob}(\lambda > 4 \mid \mathbf{Y}) \approx 0.85$.
5	$\text{Beta}(a+m,\ b+Y) = \text{Beta}(6, 11)$; 95% CI $(0.152, 0.587)$.
6	$\text{E}(\theta \mid y) = 12.8 + 0.8y$; spurious "effect" ≈ 6.0 bpm; fix with a randomised control group.
7	$\text{Gamma}(56.9,\ 0.754)$: SD 10, median 75.
8	$\text{Prob}(\sigma > 1) \approx 0.225$ vs 0.237 (ratio 0.95); $\text{Prob}(\sigma > 2) \approx 2.6 \times 10^{-5}$ vs 1.4×10^{-5} (ratio 1.77).
9	$c \approx 1.3$; that is *not* uninformative for α, β.
10	$\text{Gamma}(44.01, 50.01)$: mean 0.88, CI $(0.64, 1.16)$; Poisson fits badly; $\text{Prob}(\lambda > 1) \approx 0.18$.
11	Geometric; prior $\text{Prob}(Y \geq 7) = 1/8$; posterior $\text{Beta}(2, 6)$; PPD $= 3/13 \approx 0.231$.
12	$\lambda \mid$ data $\sim \text{Gamma}(a+7,\ b+95.9)$; mean ≈ 0.073/week.
13	Experts: $\text{Beta}(17.1, 0.9)$, $\text{Beta}(2.4, 0.6)$; posterior mixture weights $0.62/0.38$.
14	No prior is flat on both θ and the odds γ.
15	$p(\mu \mid Y) \propto e^{-(Y-\mu)^2/2}$ integrates to $\sqrt{2\pi}$: $\mu \mid Y \sim \text{N}(Y, 1)$.
16	$m(Y) \geq c(Y) \int \pi(\theta)\, \text{d}\theta = \infty$.
17	$f(\mathbf{y} \mid \theta) \geq \prod_i 0.5\phi(y_i) > 0$, then apply Exercise 16.
18	$\text{Prob}(\delta > 0 \mid z = 2) = 0.955$ (mixture) vs 0.977 (vague prior).
19	$\pi(\lambda) \propto \lambda^{-1/2}$, improper; posterior $\text{Gamma}(Y + \frac{1}{2}, 1)$, proper for every $Y \geq 0$.
20	$\pi(\lambda) \propto \lambda^{-1}$, improper; posterior $\text{Gamma}(1, Y)$, proper iff $Y > 0$.
21	Jeffreys: $\lambda \mid Y \sim \text{Gamma}(10.5, 1)$, mean 10.5, CI $(5.1, 17.7)$.
22	EB prior $\text{Beta}(5.43, 5.89)$ shrinks small counties toward 0.48 and narrows the intervals.

A five-minute primer on the tools of this chapter

Background you need

1. Conjugate pairs. A prior is *conjugate* to a likelihood if the posterior is in the same family as the prior. The ones used in this chapter:

Likelihood	Prior	Posterior
$Y \sim \text{Binomial}(n, \theta)$	$\text{Beta}(a, b)$	$\text{Beta}(a + Y,\ b + n - Y)$
$Y_i \overset{\text{iid}}{\sim} \text{Poisson}(\lambda)$	$\text{Gamma}(a, b)$	$\text{Gamma}(a + \sum Y_i,\ b + n)$
$Y_i \overset{\text{iid}}{\sim} \text{N}(\mu, \sigma^2)$, σ^2 known	$\text{N}(m, \sigma^2/k)$	$\text{N}\big(\frac{n\bar{Y}+km}{n+k}, \frac{\sigma^2}{n+k}\big)$
$Y_i \overset{\text{iid}}{\sim} \text{N}(0, \sigma^2)$	$\text{InvGamma}(a, b)$	$\text{InvGamma}(a + \frac{n}{2},\ b + \frac{\sum Y_i^2}{2})$

Reading the updates: a $\text{Beta}(a, b)$ prior acts like a earlier successes and b failures; a $\text{Gamma}(a, b)$ prior like a events in b units of exposure; a $\text{N}(m, \sigma^2/k)$ prior like k extra observations equal to m.

2. Recognising a posterior. Multiply likelihood $\times$ prior, drop every factor without the parameter, collect powers and exponents, and match a kernel: $\theta^{a-1}(1-\theta)^{b-1} \to$ Beta; $\lambda^{a-1}e^{-b\lambda} \to$ Gamma; $x^{-a-1}e^{-b/x} \to$ InvGamma; $e^{-\frac{A}{2}x^2+Bx} \to \text{N}(B/A, 1/A)$.

3. Improper priors. A prior with $\int \pi(\theta)\,\mathrm{d}\theta = \infty$ (e.g. $\pi(\theta) \propto 1$) is allowed *only if* the posterior is proper, i.e. $\int f(Y \mid \theta)\pi(\theta)\,\mathrm{d}\theta < \infty$.

4. Jeffreys prior. $\pi(\theta) \propto \sqrt{I(\theta)}$ with Fisher information $I(\theta) = -\text{E}\big[\frac{\mathrm{d}^2}{\mathrm{d}\theta^2}\log f(Y \mid \theta)\big]$. It gives the same answer whichever parameterisation you start from.

5. Mixture priors. If $\pi(\theta) = \sum_k w_k \pi_k(\theta)$ with each π_k conjugate, the posterior is $\sum_k w_k' p_k(\theta \mid Y)$ with new weights $w_k' \propto w_k\, m_k(Y)$, where $m_k(Y) = \int f(Y \mid \theta)\pi_k(\theta)\,\mathrm{d}\theta$ is the marginal likelihood under component k.

6. Empirical Bayes. Estimate the prior's hyperparameters from the data (e.g. by matching moments), then do an ordinary Bayesian analysis with that prior.

Exercise 2.1 — Frequentist coverage of a Bayesian interval

Exercise 1 (exact text from the book)

Assume $Y_1, \ldots, Y_n | \mu \overset{\text{iid}}{\sim} \text{Normal}(\mu, \sigma^2)$ where σ^2 is fixed and the unknown mean μ has prior $\mu \sim \text{Normal}(0, \sigma^2/m)$.

(a) Give a 95% posterior interval for μ.

(b) Select a value of m, and argue that for this choice your 95% posterior credible interval has frequentist coverage 0.95. That is, if you draw many samples of size n and compute the 95% interval following the formula in (a) for each sample, in the long-run 95% of the intervals will contain the true value of μ.

In plain words

Work out the posterior of a normal mean under a normal prior, write its 95% credible interval, then find the prior strength m for which this Bayesian interval also behaves like a classical 95% confidence interval.

Background you need

- **Precision** $= 1/\text{variance}$. For normal data with a normal prior, precisions add and the posterior mean is a precision-weighted average of the data mean and the prior mean.
- $\bar{Y} \sim \text{N}(\mu, \sigma^2/n)$, so $\bar{Y}$ carries precision n/σ^2.
- For $Z \sim \text{N}(0,1)$: $\text{Prob}(|Z| \leq 1.96) = 0.95$.

Key idea

The prior $\text{N}(0, \sigma^2/m)$ is worth m imaginary observations equal to 0. As $m \to 0$ the prior disappears and the Bayesian interval becomes the classical one.

Solution

(a) Step 1 — likelihood and prior as functions of μ.

$$f(\mathbf{Y} \mid \mu) \propto \exp\Big\{ -\frac{n}{2\sigma^2}(\mu - \bar{Y})^2 \Big\}, \qquad \pi(\mu) \propto \exp\Big\{ -\frac{m}{2\sigma^2}\mu^2 \Big\}.$$

Why? $\sum_i (Y_i - \mu)^2 = \sum_i (Y_i - \bar{Y})^2 + n(\bar{Y} - \mu)^2$, and the first sum does not involve μ.

(a) Step 2 — multiply and complete the square.

The exponent is $-\frac{1}{2\sigma^2}\big[n(\mu - \bar{Y})^2 + m\mu^2\big]$. Expanding, the μ^2 coefficient is $\frac{n+m}{\sigma^2}$ and the μ coefficient is $\frac{n\bar{Y}}{\sigma^2}$, so (Recall box of the primer, point 2)

$$\mu \mid \mathbf{Y} \sim \text{Normal}\left(\frac{n\bar{Y}}{n+m}, \frac{\sigma^2}{n+m} \right).$$

(a) Step 3 — the interval.

For a normal posterior, mean $\pm 1.96 \times$SD contains 95% (equal-tailed = HPD by symmetry):

$$\frac{n\bar{Y}}{n+m} \pm 1.96 \frac{\sigma}{\sqrt{n+m}}.$$

› (b) Choose $m = 0$.

The interval becomes $\bar{Y} \pm 1.96\,\sigma/\sqrt{n}$. Now think of repeated samples with the true μ fixed: $\bar{Y} \sim \mathrm{N}(\mu, \sigma^2/n)$, so $Z = \frac{\bar{Y}-\mu}{\sigma/\sqrt{n}} \sim \mathrm{N}(0,1)$ and

$$\mathrm{Prob}\Big(\mu \in \bar{Y} \pm 1.96\tfrac{\sigma}{\sqrt{n}}\Big) = \mathrm{Prob}\Big(|\bar{Y}-\mu| \le 1.96\tfrac{\sigma}{\sqrt{n}}\Big) = \mathrm{Prob}(|Z| \le 1.96) = 0.95,$$

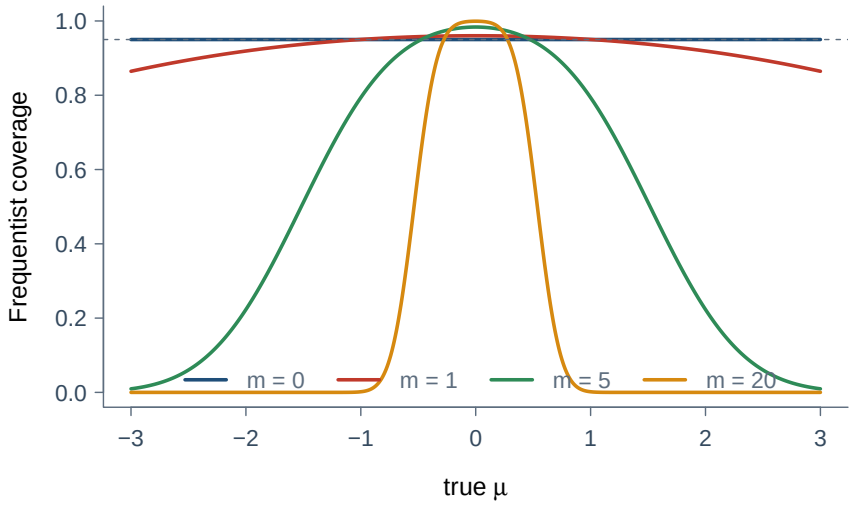
for *every* value of μ. ($m = 0$ is the limit of an infinitely vague prior — the flat prior $\pi(\mu) \propto 1$.)

› Why other m do not work.

For general m, $\frac{n\bar{Y}}{n+m} - \mu$ has mean $-\frac{m\mu}{n+m}$ (a bias toward 0) and SD $\frac{\sqrt{n}\sigma}{n+m}$. Standardising gives the coverage

$$C(\mu) = \Phi\Big(1.96\sqrt{\tfrac{n+m}{n}} + \tfrac{m\mu}{\sqrt{n}\,\sigma}\Big) - \Phi\Big(-1.96\sqrt{\tfrac{n+m}{n}} + \tfrac{m\mu}{\sqrt{n}\,\sigma}\Big).$$

For $n = 10$, $\sigma = 1$:



Near $\mu = 0$ the coverage exceeds 0.95 (the prior is right); far from 0 it collapses (the interval is pulled toward the wrong place).

Sanity check

Simulation with $m = 0$, $\mu = 1.7$, $n = 10$ and 10^5 repeated samples: 94.94% of the intervals contain μ. ✓

Answer

(a) $\dfrac{n\bar{Y}}{n+m} \pm 1.96\dfrac{\sigma}{\sqrt{n+m}}$. (b) $m = 0$: the interval becomes $\bar{Y} \pm 1.96\sigma/\sqrt{n}$, whose coverage is exactly 0.95 for every μ.

Takeaway & common mistakes

A credible interval is a statement about μ *given the data*; a confidence interval is a statement about the *procedure*. They coincide under a flat prior here.

Exercise 2.2 — Is Reggie Jackson really "Mr. October"?

Exercise 2 (exact text from the book)

The Major League Baseball player Reggie Jackson is known as "Mr. October" for his outstanding performances in the World Series (which takes place in October). Over his long career he played in 2820 regular-season games and hit 563 home runs in these games (a player can hit 0, 1, 2, … home runs in a game). He also played in 27 World Series games and hit 10 home runs in these games. Assuming uninformative conjugate priors, summarize the posterior distribution of his home-run rate in the regular season and World Series. Is there sufficient evidence to claim that he performs better in the World Series?

In plain words

Estimate his home runs per game in the regular season and in the World Series, and judge whether the World Series rate is convincingly higher.

Background you need

Poisson–gamma: if $Y_1, \ldots, Y_n \overset{\text{iid}}{\sim} \text{Poisson}(\lambda)$ and $\lambda \sim \text{Gamma}(a, b)$, then $\lambda \mid \mathbf{Y} \sim \text{Gamma}(a + \sum Y_i,\ b + n)$. Only the total count and the number of games are needed.

Key idea

Fit one Poisson–gamma model per setting; compare the two rates by drawing from both posteriors and computing their ratio (Monte Carlo).

Solution

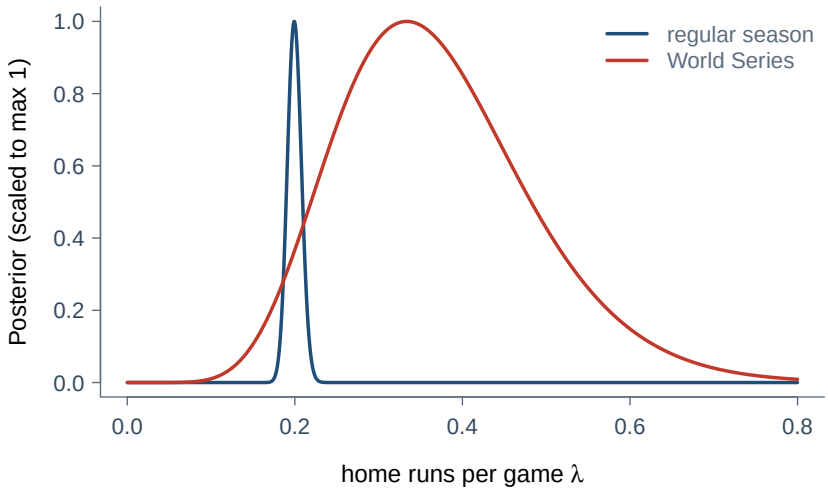
Step 1 — prior.

$\text{Gamma}(0.01, 0.01)$ for each rate: prior mean 1, variance 100 — worth only 0.01 games of data, so it is essentially uninformative.

Step 2 — posteriors.

Regular season: $\text{Gamma}(0.01 + 563,\ 0.01 + 2820)$. World Series: $\text{Gamma}(0.01 + 10,\ 0.01 + 27)$.

	Posterior	Mean $= a/b$	SD $= \sqrt{a}/b$	95% interval
Regular season	$\text{Gamma}(563.01,\ 2820.01)$	0.200	0.008	(0.184, 0.217)
World Series	$\text{Gamma}(10.01,\ 27.01)$	0.371	0.117	(0.178, 0.633)



› Step 3 — compare by simulation.

Draw 10^6 values of each rate, then count how often $\lambda_{WS} > \lambda_R$ and summarise λ_{WS}/λ_R:

$$\text{Prob}(\lambda_{WS} > \lambda_R \mid \text{data}) \approx 0.951, \qquad \frac{\lambda_{WS}}{\lambda_R} : \text{ median } 1.80,\ 95\% \text{ interval } (0.89,\ 3.19).$$

Sanity check

Posterior means are essentially the raw rates: $563/2820 = 0.200$ and $10/27 = 0.370$. ✓

Answer

Regular season ≈ 0.200 HR/game (0.184–0.217); World Series ≈ 0.371 HR/game (0.178–0.633). There is a 95% posterior probability that his World Series rate is higher, but the interval for the ratio includes 1: the evidence is **suggestive but not conclusive** (27 games is a small sample).

Takeaway & common mistakes

Beware selection bias: he earned the nickname *because* of a few spectacular games, so testing the nickname on the same data overstates the evidence.

Exercise 2.3 — Conjugate prior for the exponential rate

Exercise 3 (exact text from the book)

Assume that $Y|\theta \sim \text{Exponential}(\theta)$, i.e., $Y|\theta \sim \text{Gamma}(1, \theta)$. Find a conjugate prior distribution for θ, and derive the resulting posterior distribution.

In plain words

Find a family of priors such that, after observing an exponential waiting time, the posterior stays in the same family.

Background you need

Exponential(θ) with rate θ: $f(y \mid \theta) = \theta e^{-\theta y}$ for $y > 0$, mean $1/\theta$. Gamma kernel: $\theta^{a-1}e^{-b\theta}$.

Key idea

Look at the likelihood *as a function of* θ: $\theta^1 e^{-y\theta}$ has the Gamma shape "power of θ times exponential of θ".

Solution

› Step 1 — likelihood as a function of θ.

$f(Y \mid \theta) = \theta\, e^{-\theta Y}$.

› Step 2 — try a Gamma prior.

$\pi(\theta) \propto \theta^{a-1}e^{-b\theta}$.

› Step 3 — multiply.

$$p(\theta \mid Y) \propto \theta e^{-\theta Y} \cdot \theta^{a-1}e^{-b\theta} = \theta^{(a+1)-1}\, e^{-(b+Y)\theta}.$$

Why? Add the powers of θ: $1 + (a-1) = a = (a+1) - 1$; add the exponents: $-\theta Y - b\theta = -(b+Y)\theta$.

This is the $\text{Gamma}(a+1,\ b+Y)$ kernel, so the Gamma family is conjugate.

› Step 4 — n observations.

The likelihood becomes $\theta^n e^{-\theta \sum Y_i}$, giving $\theta \mid \mathbf{Y} \sim \text{Gamma}(a+n,\ b+\sum Y_i)$.

Sanity check

Posterior mean $\frac{a+n}{b+\sum Y_i}$. With a vague prior ($a, b \to 0$) this becomes $n/\sum Y_i = 1/\bar{Y}$, the usual estimate of a rate. ✓

Answer

The Gamma family is conjugate: $\theta \sim \text{Gamma}(a, b) \Rightarrow \theta \mid Y \sim \text{Gamma}(a+1,\ b+Y)$; for n observations, $\text{Gamma}(a+n,\ b+\sum Y_i)$.

Takeaway & common mistakes

Interpretation: a = prior number of events, b = prior total waiting time. The posterior mean is "events ÷ total time".

Exercise 2.4 — Factory injury rate

Exercise 4 (exact text from the book)

A large factory's objective is to keep the injury rate below 4 injuries per month. In the past 6 months there have been 4, 5, 2, 7, 6 and 6 injuries. Conduct a Bayesian analysis of these data and compute the posterior probability that the injury rate exceeds four injuries per month.

In plain words

From six monthly injury counts, estimate the monthly injury rate and how likely it is to exceed the target of 4.

Background you need

Poisson–gamma (primer, point 1): $\lambda \mid \mathbf{Y} \sim \text{Gamma}(a+\sum Y_t,\ b+n)$. A tail probability $\text{Prob}(\lambda > 4)$ is `pgamma(4, shape, rate, lower.tail = FALSE)` in R.

Key idea

Counts per month ⇒ Poisson likelihood; vague Gamma prior; the answer is a posterior tail area.

Solution

› Step 1 — model.

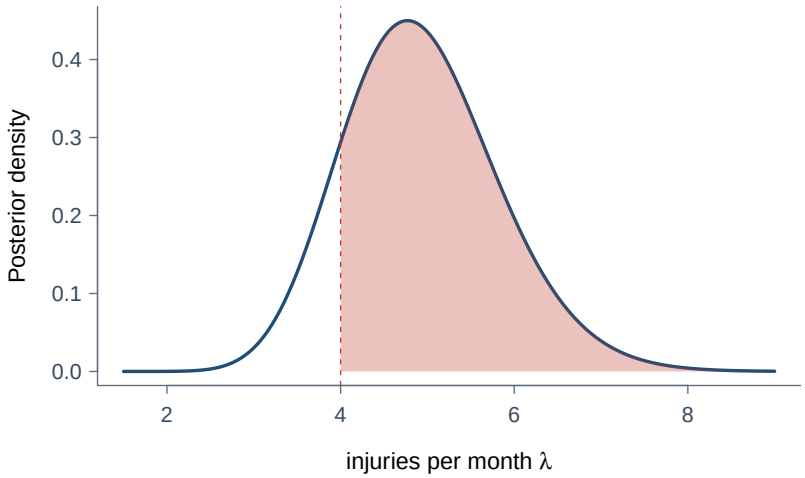
$Y_t \mid \lambda \overset{\text{iid}}{\sim} \text{Poisson}(\lambda)$, $t = 1, \ldots, 6$; prior $\lambda \sim \text{Gamma}(0.1, 0.1)$ (mean 1, variance 10: vague).

› Step 2 — sufficient statistics.

$\sum Y_t = 4+5+2+7+6+6 = 30$, $n = 6$.

› Step 3 — posterior.

$\lambda \mid \mathbf{Y} \sim \text{Gamma}(0.1+30,\ 0.1+6) = \text{Gamma}(30.1,\ 6.1)$: mean $\frac{30.1}{6.1} = 4.93$, 95% interval $(3.33,\ 6.85)$.



› Step 4 — tail probability

(shaded area): $\text{Prob}(\lambda > 4 \mid \mathbf{Y}) = 0.853$.

› Step 5 — sensitivity to the prior.

Prior	Posterior	Mean	$\text{Prob}(\lambda > 4 \mid \mathbf{Y})$
$\text{Gamma}(0.01, 0.01)$	$\text{Gamma}(30.01, 6.01)$	4.99	0.867
$\text{Gamma}(0.1, 0.1)$	$\text{Gamma}(30.1, 6.1)$	4.93	0.853
$\text{Gamma}(1, 1)$	$\text{Gamma}(31, 7)$	4.43	0.690

Sanity check

The sample mean is $30/6 = 5$ injuries per month, and the posterior mean (4.93) sits right next to it, as it should with a vague prior. ✓

Answer

$\lambda \mid \mathbf{Y} \sim \text{Gamma}(30.1, 6.1)$, posterior mean ≈ 4.9 injuries/month, and $\text{Prob}(\lambda > 4 \mid \mathbf{Y}) \approx \mathbf{0.85}$: the factory is probably missing its target.

Takeaway & common mistakes

With only six months, the prior matters: $\text{Gamma}(1, 1)$ acts like an extra month with one injury and lowers the probability to 0.69. Always report the prior.

Exercise 2.5 — Negative binomial likelihood with a beta prior

Exercise 5 (exact text from the book)

Assume that $Y|\theta \sim \text{NegBinomial}(\theta, m)$ (see Appendix A.1) and $\theta \sim \text{Beta}(a, b)$.

(a) Derive the posterior of θ.

(b) Plot the posterior of θ, and give its 95% credible interval assuming $m = 5$, $Y = 10$, and $a = b = 1$.

In plain words

Y counts the failures before the m-th success. With a beta prior on the success probability, find the posterior and evaluate it for $m = 5$, $Y = 10$.

Background you need

Appendix A.1: $\text{Prob}(Y = y \mid \theta) = \binom{y+m-1}{y}\theta^m(1-\theta)^y$, $y = 0, 1, 2, \ldots$ (failures before the m-th success). Beta kernel: $\theta^{a-1}(1-\theta)^{b-1}$.

Key idea

As a function of θ, the likelihood $\theta^m(1-\theta)^Y$ is a Beta kernel, so the Beta prior is conjugate.

Solution

(a) Step 1 — drop the constant.

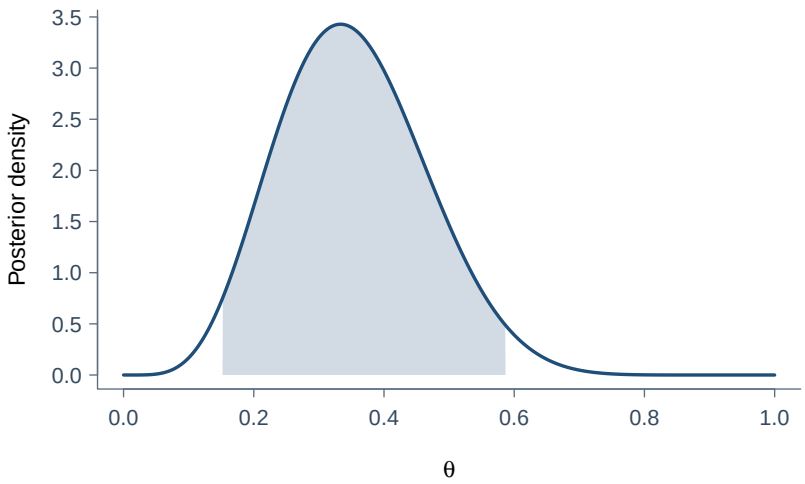
The binomial coefficient does not involve θ: $f(Y \mid \theta) \propto \theta^m(1-\theta)^Y$.

(a) Step 2 — multiply by the prior.

$$p(\theta \mid Y) \propto \theta^m(1-\theta)^Y \cdot \theta^{a-1}(1-\theta)^{b-1} = \theta^{(a+m)-1}(1-\theta)^{(b+Y)-1} \;\Rightarrow\; \theta \mid Y \sim \text{Beta}(a+m,\; b+Y).$$

(b) Numbers.

$\text{Beta}(1+5,\; 1+10) = \text{Beta}(6, 11)$: mean $\frac{6}{17} = 0.353$, 95% interval (`qbeta(c(.025,.975),6,11)`) $= (0.152,\; 0.587)$ (shaded).



Sanity check

5 successes and 10 failures: the raw success rate is $5/15 = 0.333$, close to the posterior mean 0.353 (pulled slightly toward 0.5 by the uniform prior). ✓

Answer

(a) $\theta \mid Y \sim \text{Beta}(a+m,\ b+Y)$. (b) $\text{Beta}(6, 11)$; 95% credible interval $(0.152,\ 0.587)$.

Takeaway & common mistakes

This is the same posterior as 5 successes and 10 failures under a *binomial* likelihood: the stopping rule does not affect Bayesian inference (the **likelihood principle**).

Exercise 2.6 — Heart rates and regression to the mean

Exercise 6 (exact text from the book)

The distribution of true heart rate across a population is Gaussian with mean 64 beats per minute (bpm) and standard deviation 10 bpm. An individual measurement is unbiased for the true heart rate, but has error standard deviation 5 bpm.

(a) Given that an individual has a (single) measured heart rate of y bpm, what is the posterior distribution of their true heart rate? Plot the posterior mean as a function of the observed value and comment on the regression to the mean.

(b) Consider a study designed to test a new method to treat high heart rate. To obtain a sample, they measure randomly-selected individuals from the population and include them in the study if and only if their heart rate exceeds 90 bpm. For these individuals, they apply the treatment and measure their heart rate again. The estimate of the treatment effect is the difference in mean heart rate before and after the treatment for the individuals in the study. Assuming the new method has no actual effect on heart rate, compute (using mathematical derivations or simulations) the expected difference in the mean.

(c) How could the study in (b) be redesigned to avoid regression to the mean bias?

In plain words

A noisy heart-rate reading should be pulled back toward the population average. Show this, and show how it creates a fake "treatment effect" in a study that only enrols people with high readings.

Background you need

- Normal–normal: prior $\theta \sim \mathrm{N}(m, \tau^2)$, data $y \mid \theta \sim \mathrm{N}(\theta, s^2) \Rightarrow$ posterior precision $\frac{1}{\tau^2} + \frac{1}{s^2}$ and posterior mean $w\,y + (1-w)\,m$ with $w = \frac{\tau^2}{\tau^2+s^2}$.
- **Truncated normal mean**: if $X \sim \mathrm{N}(\mu, \sigma^2)$ then $\mathrm{E}(X \mid X > c) = \mu + \sigma \frac{\phi(z)}{1-\Phi(z)}$ with $z = \frac{c-\mu}{\sigma}$.

Key idea

A high reading is partly a high true rate and partly lucky measurement error. The error does not repeat, so the second reading tends to be lower even with no treatment.

Solution

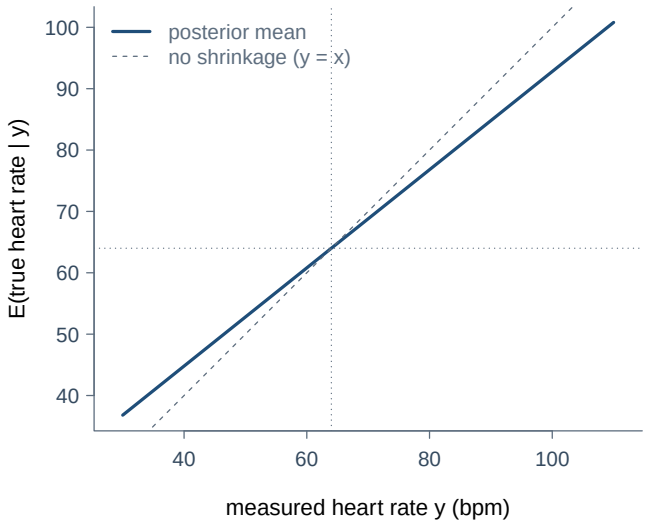
(a) Step 1 — precisions.

Prior precision $\frac{1}{100} = 0.01$; measurement precision $\frac{1}{25} = 0.04$; total $0.05 \Rightarrow$ posterior variance $\frac{1}{0.05} = 20$ (SD 4.47).

(a) Step 2 — posterior mean.

$$\mathrm{E}(\theta \mid y) = \frac{0.01 \times 64 + 0.04 \times y}{0.05} = 0.2 \times 64 + 0.8\,y = 12.8 + 0.8y.$$

So $\theta \mid y \sim \mathrm{Normal}(12.8 + 0.8y,\ 20)$.



The line has slope 0.8 (flatter than the dashed diagonal) and crosses it at 64. A reading of 100 implies a true rate of $12.8 + 80 = 92.8$; a reading of 40 implies 44.8. Extreme readings are pulled toward 64: *regression to the mean.*

› (b) Step 1 — set up.

Let $Y_1 = \theta + e_1$ (screening = "before") and $Y_2 = \theta + e_2$ ("after"; the treatment does nothing). Since e_2 has mean 0 and is independent of everything else,

$$\mathrm{E}(Y_1 - Y_2 \mid Y_1 > 90) = \mathrm{E}(Y_1 \mid Y_1 > 90) - \mathrm{E}(\theta \mid Y_1 > 90).$$

› (b) Step 2 — use part (a).

$\mathrm{E}(\theta \mid Y_1) = 12.8 + 0.8 Y_1$, so $\mathrm{E}(\theta \mid Y_1 > 90) = 12.8 + 0.8\,\mathrm{E}(Y_1 \mid Y_1 > 90)$ and

$$\mathrm{E}(Y_1 - Y_2 \mid Y_1 > 90) = 0.2\,\mathrm{E}(Y_1 \mid Y_1 > 90) - 12.8 = 0.2\big(\mathrm{E}(Y_1 \mid Y_1 > 90) - 64\big).$$

› (b) Step 3 — truncated normal mean.

Marginally $Y_1 \sim \mathrm{N}(64,\ 10^2 + 5^2 = 125)$, SD 11.18. With $z = \frac{90-64}{11.18} = 2.326$: $\phi(z) = 0.0267$, $1 - \Phi(z) = 0.0100$, so

$$\mathrm{E}(Y_1 \mid Y_1 > 90) = 64 + 11.18 \times \frac{0.0267}{0.0100} = 93.79.$$

Expected difference $= 0.2 \times (93.79 - 64) = \mathbf{5.96}$ bpm.

Sanity check

Simulation: 2×10^6 people, 20,047 enrolled, average before-minus-after $= 5.92$ bpm with a useless treatment. ✓

› (c) Redesign.

- **Randomise**: enrol people with high readings, then randomly give half a placebo. Both groups regress to the mean equally; compare the groups' *after* values.
- Or take a *fresh baseline* measurement after enrolment (independent of the screening value) and measure change from that.
- Averaging several screening readings reduces (but does not remove) the bias.

Answer

(a) $\theta \mid y \sim \text{Normal}(12.8 + 0.8y,\ 20)$. (b) $\approx$ **6** bpm apparent drop with a useless treatment. (c) Use a randomised control group (or a fresh baseline).

Takeaway & common mistakes

Selecting people on an extreme noisy value guarantees their next value is, on average, less extreme. Many "miracle cures" are regression to the mean.

Exercise 2.7 — A prior for California wildfires

Exercise 7 (exact text from the book)

Over the past 50 years California has experienced an average of $\lambda_0 = 75$ large wildfires per year. For the next 10 years you will record the number of large fires in California and then fit a Poisson/gamma model to these data. Let the rate of large fires in this future period, λ, have prior $\lambda \sim \text{Gamma}(a, b)$. Select a and b so that the prior is uninformative with prior variance around 100 and gives prior probability approximately $\text{Prob}(\lambda > \lambda_0) = 0.5$ so that the prior places equal probability on both hypotheses in the test for a change in the rate.

In plain words

Choose a Gamma prior for the future fire rate whose median is 75 (equal odds of increase or decrease) and whose variance is 100.

Background you need

$\text{Gamma}(a, b)$: mean a/b, variance a/b^2. Its median is slightly *below* its mean (right skew); R gives it as `qgamma(0.5, a, b)`.

Key idea

Two conditions, two unknowns. The variance condition gives b in terms of a; then solve the median condition numerically.

Solution

Step 1 — quick approximation using the mean.

Set mean $= 75$ and variance $= 100$: $\frac{a}{b} = 75$, $\frac{a}{b^2} = 100$. Dividing, $b = \frac{75}{100} = 0.75$ and $a = 75 \times 0.75 = 56.25$. But the median of $\text{Gamma}(56.25, 0.75)$ is a little below 75: $\text{Prob}(\lambda > 75) = 0.482$.

Step 2 — exact solution.

From $\frac{a}{b^2} = 100$: $b = \sqrt{a}/10$. Find a such that `qgamma(0.5, a, sqrt(a)/10)` $= 75$ (`uniroot`): $a = 56.91$, $b = 0.754$.

Step 3 — is it "uninformative"?

In the Poisson–gamma update, b acts like a number of prior *years* of data. Here $b = 0.75$ — less than one year — while we will collect 10 years. So the data will dominate.

Sanity check

$\text{Gamma}(56.91, 0.754)$: mean $56.91/0.754 = 75.4$, SD $\sqrt{56.91}/0.754 = 10.0$, $\text{Prob}(\lambda > 75) = 0.500$. ✓

Answer

$\lambda \sim \text{Gamma}(a = 56.9,\ b = 0.754)$: variance 100 (SD 10), median 75. The simpler $\text{Gamma}(56.25, 0.75)$ (mean 75) gives $\text{Prob}(\lambda > 75) = 0.48$, close enough to "approximately 0.5".

Takeaway & common mistakes

Judge "uninformative" *relative to the data*: a prior worth 0.75 years is weak against 10 years of counts, even though an SD of 10 fires is not vague in absolute terms.

Exercise 2.8 — Is the measurement error too large?

Exercise 8 (exact text from the book)

An assembly line relies on accurate measurements from an image-recognition algorithm at the first stage of the process. It is known that the algorithm is unbiased, so assume that measurements follow a normal distribution with mean zero, $Y_i|\sigma^2 \overset{\text{iid}}{\sim} \text{Normal}(0, \sigma^2)$. Some errors are permissible, but if σ exceeds the threshold c, then the algorithm must be replaced. You make $n = 20$ measurements and observe $\sum_{i=1}^n Y_i = -2$ and $\sum_{i=1}^n Y_i^2 = 15$ and conduct a Bayesian analysis with $\text{InvGamma}(a, b)$ prior. Compute the posterior probability that $\sigma > c$ for:

(a) $c = 1$ and $a = b = 0.1$

(b) $c = 1$ and $a = b = 1.0$

(c) $c = 2$ and $a = b = 0.1$

(d) $c = 2$ and $a = b = 1.0$

For each c, compute the ratio of probabilities for the two priors (i.e. $a = b = 0.1$ and $a = b = 1.0$). Which, if any, of the results are sensitive to the prior?

In plain words

Estimate the error SD from 20 measurements with known mean 0, and compute the probability it exceeds a threshold under two priors.

Background you need

- $\text{InvGamma}(a, b)$ kernel in x: $x^{-a-1}e^{-b/x}$. If $\sigma^2 \sim \text{InvGamma}(A, B)$ then $1/\sigma^2 \sim \text{Gamma}(A, B)$.
- $\sigma > c \iff \sigma^2 > c^2 \iff 1/\sigma^2 < 1/c^2$ (all quantities positive).

Key idea

Known mean, unknown variance ⇒ inverse-gamma posterior. Convert the question about σ into a question about the precision $1/\sigma^2$, which is Gamma, and use `pgamma`.

Solution

› Step 1 — likelihood.

$$f(\mathbf{Y} \mid \sigma^2) = \prod_{i=1}^{20} \frac{1}{\sqrt{2\pi\sigma^2}} e^{-Y_i^2/(2\sigma^2)} \propto (\sigma^2)^{-10} \exp\left\{ -\frac{\sum Y_i^2}{2\sigma^2} \right\} = (\sigma^2)^{-10} e^{-7.5/\sigma^2}.$$

Why? $(\sigma^2)^{-1/2}$ twenty times gives $(\sigma^2)^{-10}$; $\sum Y_i^2/2 = 15/2 = 7.5$.

$\sum Y_i = -2$ is *not needed*: the mean is known to be 0, so only $\sum Y_i^2$ enters.

› Step 2 — posterior.

$$p(\sigma^2 \mid \mathbf{Y}) \propto (\sigma^2)^{-10} e^{-7.5/\sigma^2} \cdot (\sigma^2)^{-a-1} e^{-b/\sigma^2} = (\sigma^2)^{-(a+10)-1} e^{-(b+7.5)/\sigma^2},$$

so $\sigma^2 \mid \mathbf{Y} \sim \text{InvGamma}(a + 10,\ b + 7.5)$.

› Step 3 — the probability.

$\text{Prob}(\sigma > c \mid \mathbf{Y}) = \text{Prob}(1/\sigma^2 < 1/c^2 \mid \mathbf{Y}) =$ `pgamma(1/c2, a + 10, b + 7.5)`.

	$a = b = 0.1$	$a = b = 1$	ratio
$c = 1$	0.2250	0.2366	0.95
$c = 2$	2.56×10^{-5}	1.45×10^{-5}	1.77

Sanity check

The sample estimate $\hat{\sigma}^2 = \frac{15}{20} = 0.75$ ($\hat{\sigma} = 0.87$) is below 1, so $\text{Prob}(\sigma > 1) \approx 0.23$ (less than half) and $\text{Prob}(\sigma > 2)$ tiny both make sense. ✓

Answer

(a) 0.225 (b) 0.237 (c) 2.6×10^{-5} (d) 1.4×10^{-5}.
For $c = 1$: **not** sensitive (ratio 0.95). For $c = 2$: **relatively** sensitive (ratio 1.77) in the extreme tail, but both probabilities are negligible, so the *decision* (keep the algorithm) is the same.

Takeaway & common mistakes

Tail probabilities far from the data are always the most prior-sensitive quantities. Judge sensitivity by whether it changes the decision.

Exercise 2.9 — Induced prior in logistic regression

Exercise 9 (exact text from the book)

In simple logistic regression, the probability of a success is regressed onto a single covariate X as

$$p(X) = \frac{\exp(\alpha + X\beta)}{1 + \exp(\alpha + X\beta)}. \qquad (2.51)$$

Assuming the covariate is distributed as $X \sim \text{Normal}(0, 1)$ and the parameters have priors $\alpha \sim \text{Normal}(0, c^2)$ and $\beta \sim \text{Normal}(0, c^2)$, find a prior standard deviation c so that the induced prior on the success probability $p(X)$ is roughly $\text{Uniform}(0, 1)$. That is, for a given c generate S (say $S = 1{,}000{,}000$) samples $(X^{(s)}, \alpha^{(s)}, \beta^{(s)})$ for $s = 1, \ldots, S$, compute the S success probabilities, and make a histogram of the S probabilities. Repeat this for several c until the histogram is approximately uniform. Report the final value of c and the resulting histogram. Would you call this an uninformative prior for the regression coefficients α and β?

In plain words

A prior on the regression coefficients implies a prior on the probability $p(X)$. Find the prior SD c that makes this implied prior flat.

Background you need

- If $p \sim \text{Uniform}(0, 1)$, then $\text{logit}(p) = \log \frac{p}{1-p}$ has the standard logistic distribution, variance $\pi^2/3 \approx 3.29$.
- For independent α, β, X with means 0: $\text{Var}(\alpha + X\beta) = \text{Var}(\alpha) + \text{E}(X^2)\text{Var}(\beta) = c^2 + c^2 = 2c^2$.

Key idea

First guess from matching variances, $2c^2 \approx 3.29 \Rightarrow c \approx 1.28$; then refine by simulation, measuring flatness with the Kolmogorov–Smirnov (KS) distance to the uniform.

Solution

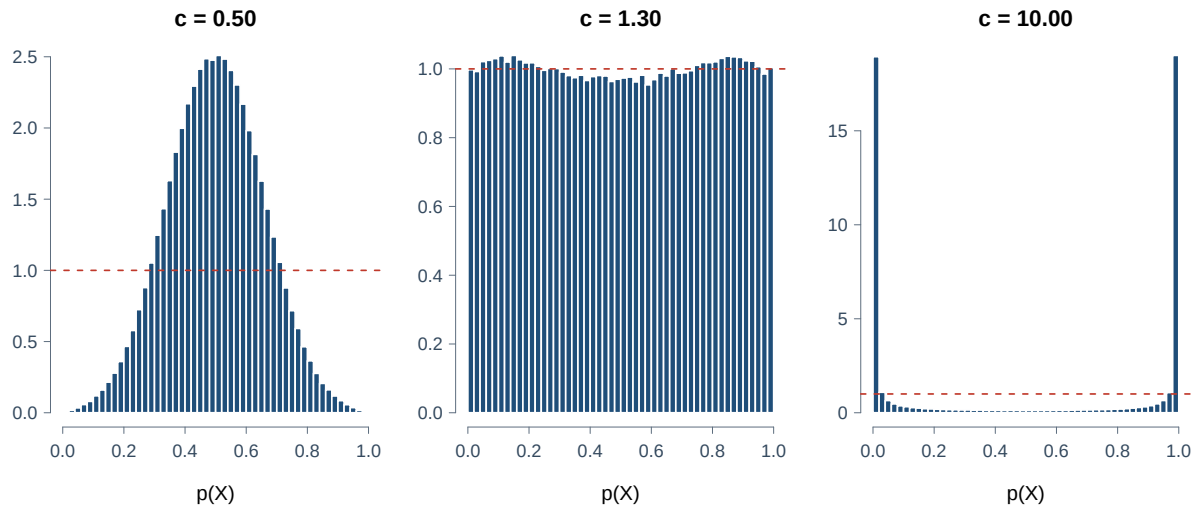
Step 1 — simulate the induced prior for a given c.

```
induced_p ← function(c, S) plogis(rnorm(S, 0, c) + rnorm(S) * rnorm(S, 0, c))
```

(`plogis` is the inverse-logit $\frac{e^\eta}{1+e^\eta}$; the three `rnorm` calls draw α, X and β.)

Step 2 — search over c.

For $c = 0.50, 0.55, \ldots, 3.00$ compute the KS distance $\max_u |\hat{F}(u) - u|$ between the simulated probabilities and the uniform. The smallest distance is at $c = 1.30$; with $S = 10^6$ the KS distance is 0.005 (essentially flat).



Left: $c = 0.5$ piles up near 0.5. Middle: $c = 1.3$ is flat. Right: the "vague" $c = 10$ pushes almost all mass to $p \approx 0$ or $p \approx 1$.

› Step 3 — uninformative for α, β?

No. With $c = 1.3$, $\text{Prob}(|\beta| > 2.6) = \text{Prob}(|Z| > 2) \approx 0.05$: odds ratios above $e^{2.6} \approx 13$ per SD of X are considered unlikely. That is a real (though sensible) restriction on the coefficients.

Sanity check

The simulated best value $c = 1.30$ agrees with the back-of-envelope variance match $c \approx 1.28$. ✓

Answer

$c \approx \mathbf{1.3}$ gives an approximately $\text{Uniform}(0, 1)$ induced prior on $p(X)$. It is **not** uninformative for α and β; it is mildly informative on the coefficient scale but flat on the probability scale.

Takeaway & common mistakes

"Uninformative" depends on the scale. A prior that looks vague for coefficients ($c = 10$) is very *informative* about probabilities. Always check what a prior implies on the scale you care about.

Exercise 2.10 — Adverse side effects at the Mayo Clinic

Exercise 10 (exact text from the book)

The Mayo Clinic conducted a study of $n = 50$ patients followed for one year on a new medication and found that 30 patients experienced no adverse side effects (ASE), 12 experienced one ASE, 6 experienced two ASEs and 2 experienced ten ASEs.

(a) Derive the posterior of λ for the Poisson/gamma model $Y_1, \ldots, Y_n | \lambda \overset{\text{iid}}{\sim} \text{Poisson}(\lambda)$ and $\lambda \sim \text{Gamma}(a, b)$.

(b) Use the Poisson/gamma model with $a = b = 0.01$ to study the rate of adverse events. Plot the posterior and give the posterior mean and 95% credible interval.

(c) Repeat this analysis with $\text{Gamma}(0.1, 0.1)$ and $\text{Gamma}(1, 1)$ priors and discuss sensitivity to the prior.

(d) Plot the data versus the $\text{Poisson}(\hat{\lambda})$ PMF, where $\hat{\lambda}$ is the posterior mean of λ from part (b). Does the Poisson likelihood fit well?

(e) The current medication is thought to have around one adverse side effect per year. What is the posterior probability that this new medication has a higher side effect rate than the previous medication? Are the results sensitive to the prior?

In plain words

Estimate the yearly rate of side effects from 50 patients, check prior sensitivity and model fit, and compare with a rate of 1 per year.

Background you need

For a Poisson distribution, the variance equals the mean. If the data's variance is much larger than its mean, the data are **over-dispersed** and a Poisson model is too narrow.

Key idea

Poisson–gamma conjugacy for (a)–(c) and (e); for (d) compare observed frequencies with $\text{Poisson}(\hat{\lambda})$ probabilities.

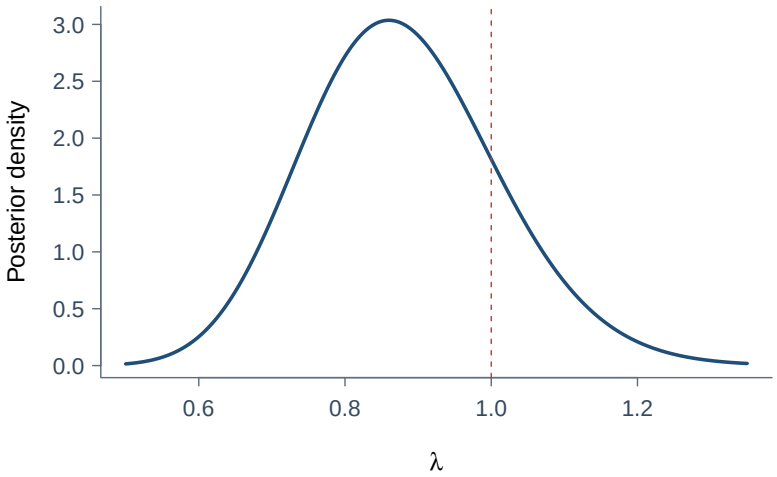
Solution

(a) Derivation.

$$p(\lambda \mid \mathbf{Y}) \propto \prod_{i=1}^{n} \frac{e^{-\lambda}\lambda^{Y_i}}{Y_i!} \cdot \lambda^{a-1}e^{-b\lambda} \propto \lambda^{a+\sum Y_i - 1}e^{-(b+n)\lambda} \;\Rightarrow\; \lambda \mid \mathbf{Y} \sim \text{Gamma}\Big(a + \sum Y_i,\; b + n\Big).$$

(b) Numbers.

$\sum Y_i = 0 \times 30 + 1 \times 12 + 2 \times 6 + 10 \times 2 = 44$, $n = 50$: $\lambda \mid \mathbf{Y} \sim \text{Gamma}(44.01,\ 50.01)$, mean 0.880, 95% interval $(0.639,\ 1.158)$.

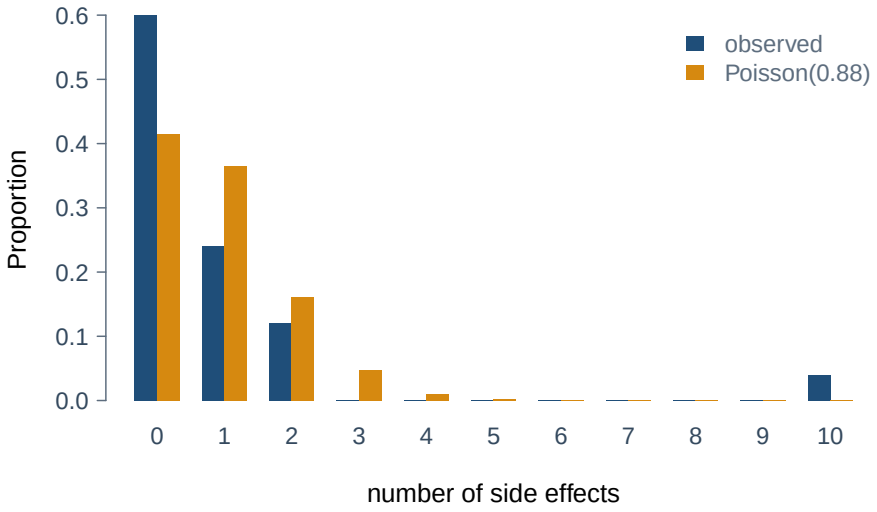


› (c) Sensitivity.

Prior	Posterior	Mean	95% interval	$\text{Prob}(\lambda > 1)$
$\text{Gamma}(0.01, 0.01)$	$\text{Gamma}(44.01, 50.01)$	0.880	(0.639, 1.158)	0.180
$\text{Gamma}(0.1, 0.1)$	$\text{Gamma}(44.1, 50.1)$	0.880	(0.640, 1.158)	0.180
$\text{Gamma}(1, 1)$	$\text{Gamma}(45, 51)$	0.882	(0.644, 1.158)	0.182

No meaningful sensitivity: 44 events in 50 patient-years swamp priors worth at most 1 event and 1 year.

› (d) Fit.



The fit is **poor**. $\text{Poisson}(0.88)$ predicts $50 \times e^{-0.88} = 20.7$ patients with no events (observed: 30), and $\text{Prob}(Y \geq 10) = 3.5 \times 10^{-8}$, yet 2 of 50 patients had ten events. The sample variance is 4.03, about 4.6 times the mean 0.88: strong **over-dispersion**.

› (e)

$\text{Prob}(\lambda > 1 \mid \mathbf{Y}) \approx 0.18$ under all three priors: not sensitive to the prior.

Answer

(a) $\text{Gamma}(a + \sum Y_i,\ b + n)$. (b) $\text{Gamma}(44.01, 50.01)$: mean 0.88, 95% CI $(0.64, 1.16)$. (c) Insensitive.
(d) Poor fit: too many zeros and two extreme patients (over-dispersion). (e) $\text{Prob}(\lambda > 1 \mid \mathbf{Y}) \approx 0.18$, insensitive to the prior — but *sensitive to the model.*

Takeaway & common mistakes

Prior sensitivity and model sensitivity are different things. Here the Poisson model is wrong and its interval is far too narrow; a negative binomial would be more honest.

Exercise 2.11 — Days until a smoker relapses

Exercise 11 (exact text from the book)

A smoker is trying to quit and their goal is to abstain for one week because studies have shown that this is the most difficult period. Denote Y as the number of days before the smoker relapses, and θ as the probability that they relapse on any given day of the study.

(a) Argue that a negative binomial distribution (Appendix A.1) for $Y|\theta$ is reasonable. What are the most important assumptions being made, and are they valid for this smoking cessation attempt?

(b) Assuming a negative binomial likelihood and $\text{Uniform}(0, 1)$ prior for θ, what is the prior probability the smoker will be smoke-free for at least 7 days?

(c) Plot the posterior distribution of θ assuming $Y = 5$.

(d) Given this attempt resulted in $Y = 5$ and that θ is the same in the second attempt, what is the posterior predictive probability the second attempt will result in at least 7 smoke-free days?

In plain words

Each day the smoker relapses with probability θ. Model the number of smoke-free days, update with an attempt that lasted 5 days, and predict the next attempt.

Background you need

- $\text{NegBinomial}(\theta, m)$ (Appendix A.1) counts failures before the m-th success. With $m = 1$ it is the **geometric**: $\text{Prob}(Y = y \mid \theta) = \theta(1 - \theta)^y$.
- "At least k failures before the first success" means the first k trials all failed: $\text{Prob}(Y \geq k \mid \theta) = (1 - \theta)^k$.
- $\int_0^1 \theta^{a-1}(1 - \theta)^{b-1}\, \mathrm{d}\theta = B(a, b)$.

Key idea

Each day is a trial; "relapse" is the "success". Smoke-free days are the failures before the first success.

Solution

(a) Justification and assumptions.

Counting days until the first event in a sequence of trials is exactly a negative binomial with $m = 1$. Assumptions:

(i) the relapse probability θ is the **same every day** — doubtful: cravings are worst early on, so the daily risk probably *decreases*;

(ii) days are **independent** — doubtful: a stressful day raises the next day's risk too;

(iii) time is counted in whole days.

Still a reasonable first approximation.

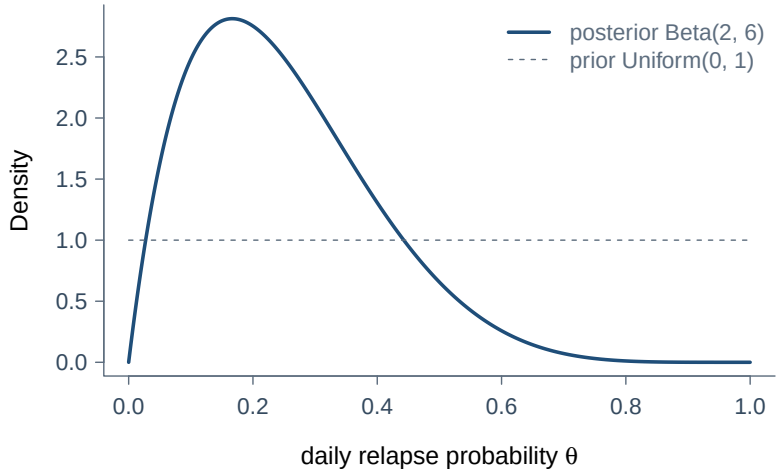
(b) Prior predictive probability.

$$\text{Prob}(Y \geq 7) = \int_0^1 \text{Prob}(Y \geq 7 \mid \theta)\, \pi(\theta)\, \mathrm{d}\theta = \int_0^1 (1 - \theta)^7\, \mathrm{d}\theta = \Big[-\frac{(1 - \theta)^8}{8} \Big]_0^1 = \frac{1}{8} = 0.125.$$

› (c) Posterior with $Y = 5$.

$$p(\theta \mid Y = 5) \propto \theta(1-\theta)^5 \cdot 1 = \theta^{2-1}(1-\theta)^{6-1} \;\Rightarrow\; \theta \mid Y \sim \text{Beta}(2, 6),$$

mean $\frac{2}{8} = 0.25$, 95% interval $(0.037,\ 0.579)$.



› (d) Posterior predictive.

$$\text{Prob}(Y^* \geq 7 \mid Y = 5) = \int_0^1 (1-\theta)^7 \frac{\theta(1-\theta)^5}{B(2,6)}\, \mathrm{d}\theta = \frac{B(2,13)}{B(2,6)}.$$

With $B(a,b) = \frac{(a-1)!(b-1)!}{(a+b-1)!}$:

$$\frac{B(2,13)}{B(2,6)} = \frac{1!\,12!/14!}{1!\,5!/7!} = \frac{1/(13\cdot 14)}{1/(6\cdot 7)} = \frac{6\cdot 7}{13\cdot 14} = \frac{42}{182} = \frac{3}{13} \approx 0.231.$$

Sanity check

Monte Carlo: draw $\theta \sim \text{Beta}(2,6)$, then $Y^* \sim \text{Geometric}(\theta)$; the fraction with $Y^* \geq 7$ is 0.2307. ✓

Answer

(a) Geometric $=$ NegBinomial$(\theta, 1)$; assumes a constant, independent daily relapse risk (doubtful).
(b) $1/8 = 0.125$. (c) Beta$(2,6)$. (d) $3/13 \approx \mathbf{0.231}$.

Takeaway & common mistakes

Plugging in $\hat{\theta} = 0.25$ would give $0.75^7 = 0.133$, understating the chance of a good week because it ignores the possibility of a small θ.

Exercise 2.12 — Censored relapse times

Exercise 12 (exact text from the book)

A smoking cessation study followed $n = 20$ patients until they relapsed to smoking. The trial terminated after six weeks and so many patients did not relapse during the observation period; i.e., their responses was censored. Seven patients relapsed during the observation period at times 0.5, 0.9, 1.1, 2.4, 3.8, 4.0 and 5.2 weeks, and the rest were censored at 6 weeks. Assume the times to event $Y_1, \ldots, Y_n$ follow the survival analysis model $Y_i|\lambda \overset{\text{iid}}{\sim} \text{Exponential}(\lambda)$ with prior $\lambda \sim \text{Gamma}(a, b)$.

(a) Show that the posterior distribution of λ is a gamma distribution (hint: the likelihood contribution of a censored observation is $\text{Prob}(Y_i > 6|\lambda) = \exp(-6\lambda)$).

(b) Compute the posterior mean and 95% interval for λ for several values of a and b. Are the results sensitive to the prior?

In plain words

Seven patients relapsed at known times; thirteen were still smoke-free when the study ended. Use both kinds of information to estimate the relapse rate.

Background you need

For Exponential(λ): density $\lambda e^{-\lambda y}$ and survival function $\text{Prob}(Y > t) = e^{-\lambda t}$.

Key idea

An observed event contributes its density; a censored patient contributes the probability of surviving past 6 weeks. The posterior then depends only on the number of events d and the total time at risk T.

Solution

(a) Step 1 — likelihood.

$$L(\lambda) = \underbrace{\prod_{i=1}^{7} \lambda e^{-\lambda y_i}}_{\text{7 relapses}} \times \underbrace{\prod_{j=1}^{13} e^{-6\lambda}}_{\text{13 censored}} = \lambda^7 \exp\Big\{ -\lambda\Big(\sum_{i=1}^{7} y_i + 13 \times 6\Big)\Big\}.$$

(a) Step 2 — total time at risk.

$\sum y_i = 0.5 + 0.9 + 1.1 + 2.4 + 3.8 + 4.0 + 5.2 = 17.9$ and $13 \times 6 = 78$, so $T = 95.9$ patient-weeks and $L(\lambda) = \lambda^7 e^{-95.9\lambda}$.

(a) Step 3 — posterior.

$$p(\lambda \mid \text{data}) \propto \lambda^7 e^{-95.9\lambda} \cdot \lambda^{a-1} e^{-b\lambda} = \lambda^{(a+7)-1} e^{-(b+95.9)\lambda} \;\Rightarrow\; \lambda \mid \text{data} \sim \text{Gamma}(a + 7,\ b + 95.9).$$

(b) Several priors.

Prior	Posterior	Mean (per week)	95% interval
Gamma(0.01, 0.01)	Gamma(7.01, 95.91)	0.073	(0.029, 0.136)
Gamma(0.1, 0.1)	Gamma(7.1, 96.0)	0.074	(0.030, 0.138)
Gamma(1, 1)	Gamma(8, 96.9)	0.083	(0.036, 0.149)
Gamma(2, 10)	Gamma(9, 105.9)	0.085	(0.039, 0.149)

The vague priors agree; priors that add one or two pseudo-events move the mean by about 15%, because there are only 7 real events.

Sanity check

Classical estimate: events ÷ time at risk $= 7/95.9 = 0.073$ per week — the same as the vague-prior posterior mean. ✓

Answer

(a) $\lambda \mid \text{data} \sim \text{Gamma}(a + 7,\ b + 95.9)$. (b) Vague priors: mean ≈ 0.073 relapses/week (mean time to relapse ≈ 14 weeks), 95% CI $\approx (0.03, 0.14)$; mildly sensitive because there are only 7 events.

Takeaway & common mistakes

Never drop censored observations. Using only the 7 relapse times gives a rate of $1/2.56 = 0.39$ per week — five times too high.

Exercise 2.13 — Mixture-of-experts prior for an alarm

Exercise 13 (exact text from the book)

You are designing a very small experiment to determine the sensitivity of a new security alarm system. You will simulate five robbery attempts and record the number of these attempts that trigger the alarm. Because the dataset will be small, you ask two experts for their opinion. One expects the alarm probability to be 0.95 with standard deviation 0.05, and the other expects 0.80 with standard deviation 0.20.

(a) Translate these two priors into beta PDFs, and plot the two beta PDFs and the corresponding mixture of experts prior with equal weight given to each expert.

(b) Now you conduct the experiment, and the alarm is triggered in every simulated robbery. Plot the posterior of the alarm probability under a uniform prior, each experts' prior, and the mixture of experts prior.

In plain words

Turn two experts' (mean, SD) opinions into beta priors, average them, and update all priors with 5 alarms out of 5 attempts.

Background you need

- **Beta moment matching**: for mean μ and variance v, $a+b=\frac{\mu(1-\mu)}{v}-1$, $a=\mu(a+b)$, $b=(1-\mu)(a+b)$.
- **Marginal likelihood** of Y successes in n trials under $\text{Beta}(a,b)$: $m(Y)=\binom{n}{Y}\frac{B(a+Y,b+n-Y)}{B(a,b)}$.
- Mixture update: primer, point 5.

Key idea

Each expert's prior updates in the usual conjugate way; the mixture weights update in proportion to how well each expert predicted the data.

Solution

(a) Expert 1

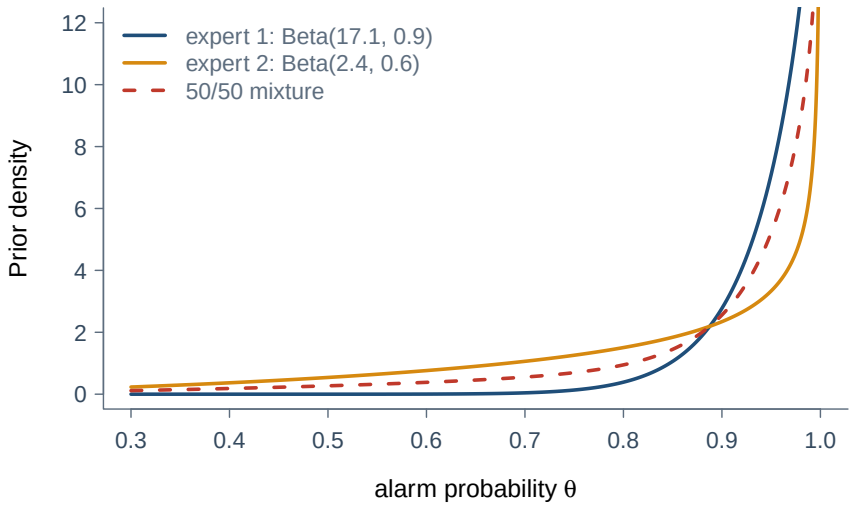
(mean 0.95, SD 0.05, variance 0.0025): $a+b=\frac{0.95\times0.05}{0.0025}-1=19-1=18$, so $a=0.95\times18=17.1$, $b=0.05\times18=0.9$: $\text{Beta}(17.1,\ 0.9)$.

(a) Expert 2

(mean 0.80, SD 0.20, variance 0.04): $a+b=\frac{0.8\times0.2}{0.04}-1=4-1=3$, so $a=2.4$, $b=0.6$: $\text{Beta}(2.4,\ 0.6)$.

(a) Mixture.

$\pi(\theta)=0.5\,\text{Beta}(\theta;17.1,0.9)+0.5\,\text{Beta}(\theta;2.4,0.6)$.



› (b) Step 1 — each conjugate posterior

($Y = 5$, $n = 5$, add 5 to a, 0 to b): uniform $\to \text{Beta}(6, 1)$ (mean 0.857); expert 1 $\to \text{Beta}(22.1, 0.9)$ (mean 0.961); expert 2 $\to \text{Beta}(7.4, 0.6)$ (mean 0.925).

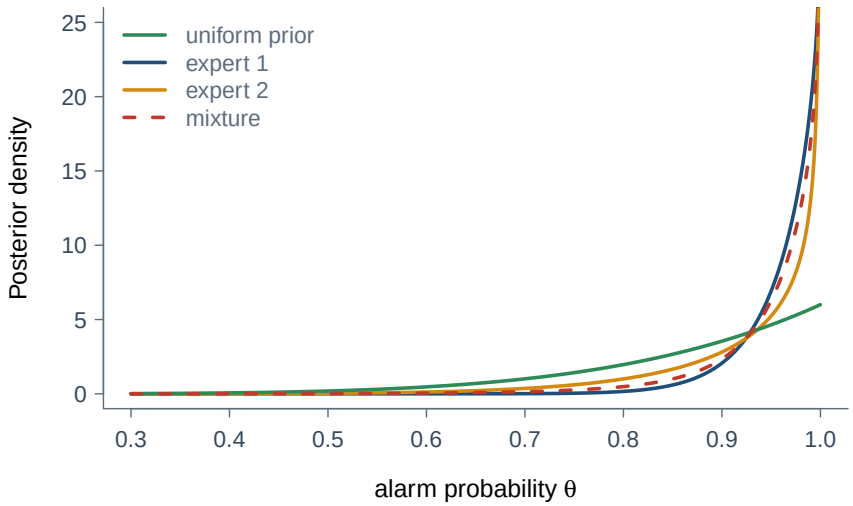
› (b) Step 2 — updated mixture weights.

$$w_1' \propto 0.5\,\frac{B(22.1, 0.9)}{B(17.1, 0.9)}, \qquad w_2' \propto 0.5\,\frac{B(7.4, 0.6)}{B(2.4, 0.6)} \quad \Rightarrow \quad w_1' = 0.617, \;\; w_2' = 0.383.$$

(The binomial coefficient $\binom{5}{5} = 1$ is common to both and cancels.)

› (b) Step 3 — mixture posterior.

$0.617\,\text{Beta}(22.1, 0.9) + 0.383\,\text{Beta}(7.4, 0.6)$, mean $0.617 \times 0.961 + 0.383 \times 0.925 = 0.947$.



Sanity check

Expert 1 predicted 5/5 more confidently (prior mean 0.95), so the data should shift weight toward expert 1 — and they do (50% → 62%). ✓

Answer

(a) Expert 1 = $\text{Beta}(17.1, 0.9)$, expert 2 = $\text{Beta}(2.4, 0.6)$, mixture = their 50/50 average.
(b) Posteriors: uniform $\text{Beta}(6, 1)$; expert 1 $\text{Beta}(22.1, 0.9)$; expert 2 $\text{Beta}(7.4, 0.6)$; mixture $0.617\,\text{Beta}(22.1, 0.9) + 0.383\,\text{Beta}(7.4, 0.6)$ (mean 0.947).

Takeaway & common mistakes

A mixture prior is a principled way to combine disagreeing experts: the data decide whom to trust.

Exercise 2.14 — Probability or odds: which scale is "flat"?

Exercise 14 (exact text from the book)

Let θ be the binomial success probability and $\gamma = \theta/(1-\theta)$ be the corresponding odds of a success. Use Monte Carlo simulation (you can discard samples with $\gamma > 100$) to explore the effects of parameterization on prior choice in this context.

(a) If $\theta \sim \text{Uniform}(0, 1)$, what is the induced prior on γ?

(b) If $\theta \sim \text{Beta}(0.5, 0.5)$, what is the induced prior on γ?

(c) If $\gamma \sim \text{Uniform}(0, 100)$, what is the induced prior on θ?

(d) If $\gamma \sim \text{Gamma}(1, 1)$, what is the induced prior on θ?

(e) Would you say any of these priors are simultaneously uninformative for both θ and γ?

In plain words

Put a prior on the probability θ and see what it says about the odds γ, and vice versa.

Background you need

Change of variables: if $\gamma = g(\theta)$ is monotone with inverse $\theta = h(\gamma)$, then $f_\gamma(\gamma) = f_\theta\big(h(\gamma)\big)\,|h'(\gamma)|$. Here $\theta = h(\gamma) = \frac{\gamma}{1+\gamma}$, $h'(\gamma) = \frac{1}{(1+\gamma)^2}$, and $1-\theta = \frac{1}{1+\gamma}$. In the other direction $\gamma = \frac{\theta}{1-\theta}$ and $\frac{d\gamma}{d\theta} = \frac{1}{(1-\theta)^2}$.

Key idea

Monte Carlo: draw from the prior on one scale and transform each draw. The change-of-variables formula gives the exact curves to overlay.

Solution

(a) $\theta \sim \text{Uniform}(0, 1)$.

$f_\theta = 1$, so $f_\gamma(\gamma) = 1 \cdot \frac{1}{(1+\gamma)^2}$ for $\gamma > 0$. Decreasing, median 1 (since $\theta = 0.5 \iff \gamma = 1$), very heavy tail ($\text{E}\gamma = \infty$).

(b) $\theta \sim \text{Beta}(0.5, 0.5)$.

$f_\theta(\theta) = \frac{1}{\pi}\theta^{-1/2}(1-\theta)^{-1/2}$. Substitute $\theta = \frac{\gamma}{1+\gamma}$, $1-\theta = \frac{1}{1+\gamma}$:

$$f_\gamma(\gamma) = \frac{1}{\pi}\Big(\frac{\gamma}{1+\gamma}\Big)^{-1/2}\Big(\frac{1}{1+\gamma}\Big)^{-1/2}\frac{1}{(1+\gamma)^2} = \frac{1}{\pi}\gamma^{-1/2}(1+\gamma)\frac{1}{(1+\gamma)^2} = \frac{1}{\pi\sqrt{\gamma}\,(1+\gamma)}.$$

An infinite spike at 0 plus a heavy tail.

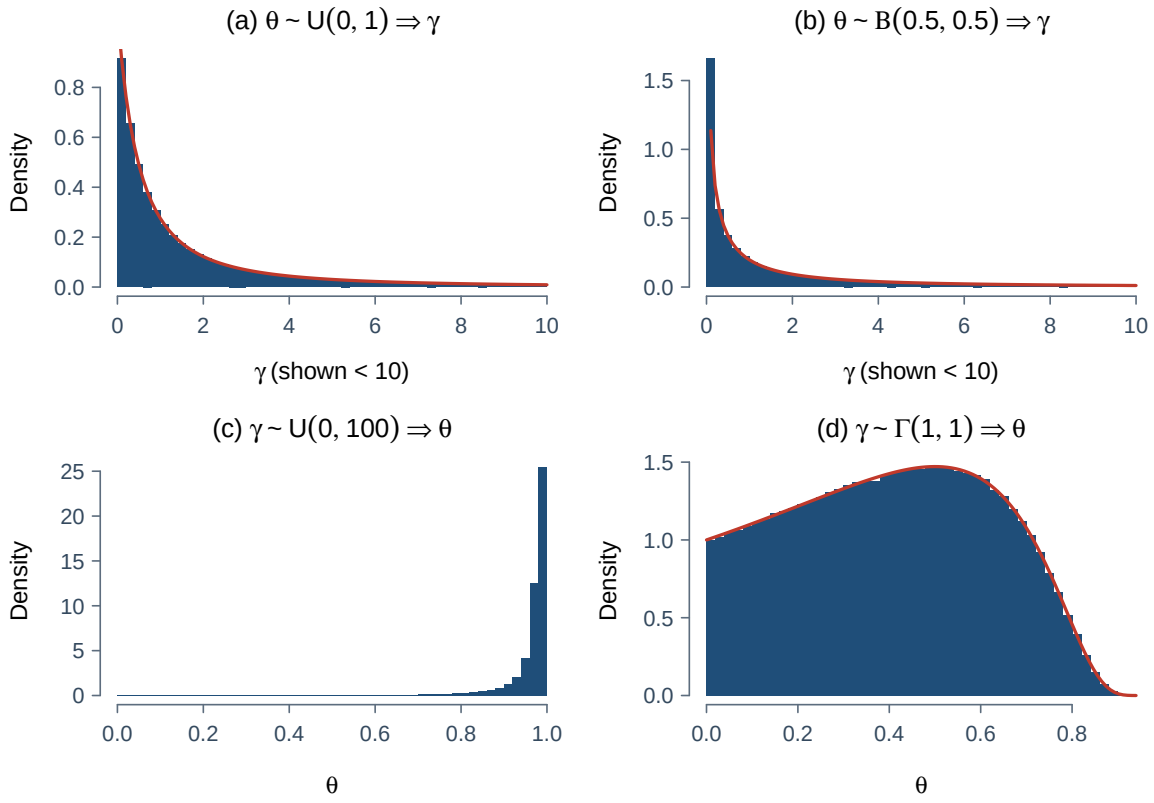
(c) $\gamma \sim \text{Uniform}(0, 100)$.

$f_\theta(\theta) = \frac{1}{100} \cdot \frac{1}{(1-\theta)^2}$ for $0 < \theta < \frac{100}{101}$. Almost all mass near 1: $\text{Prob}(\theta > 0.9) = \text{Prob}(\gamma > 9) = \frac{91}{100} = 0.91$.

(d) $\gamma \sim \text{Gamma}(1, 1) = \text{Exponential}(1)$.

$f_\theta(\theta) = e^{-\theta/(1-\theta)}\frac{1}{(1-\theta)^2}$ on $(0, 1)$: hump-shaped, falling to 0 near 1 ($\text{Prob}(\theta > 0.9) = e^{-9} \approx 0.0001$).

Simulated histograms (10^6 draws, $\gamma > 100$ discarded) with exact densities in red:



(e)

None is flat on both scales. A flat θ gives a strongly skewed γ; a flat γ pushes θ toward 1. Because $\gamma = \theta/(1-\theta)$ is nonlinear, flatness on one scale implies non-flatness on the other.

Sanity check

Each red curve lies on top of its simulated histogram. ✓

Answer

(a) $\frac{1}{(1+\gamma)^2}$ (b) $\frac{1}{\pi\sqrt{\gamma}(1+\gamma)}$ (c) $\frac{1}{100(1-\theta)^2}$ on $(0, \frac{100}{101})$ (d) $\frac{e^{-\theta/(1-\theta)}}{(1-\theta)^2}$
(e) **No** prior is uninformative on both scales.

Takeaway & common mistakes

This motivates the **Jeffreys prior**: not flat, but giving the same answer whichever parameterisation you start from. The $\mathrm{Beta}(0.5, 0.5)$ in (b) is the binomial Jeffreys prior.

Exercise 2.15 — A flat prior can give a proper posterior

Exercise 15 (exact text from the book)

Say $Y|\mu \sim \text{Normal}(\mu, 1)$ and μ has improper prior $\pi(\mu) = 1$ for all μ. Prove that the posterior distribution of μ is a proper PDF.

In plain words

The flat prior does not integrate to 1; show that the posterior still does.

Background you need

A posterior is proper iff the marginal $m(Y) = \int f(Y \mid \mu)\pi(\mu)\,\mathrm{d}\mu$ is finite. Gaussian integral: $\int_{-\infty}^{\infty} e^{-(\mu-Y)^2/2}\,\mathrm{d}\mu = \sqrt{2\pi}$.

Key idea

As a function of μ, the normal density $f(Y \mid \mu)$ has exactly the shape of a $\mathrm{N}(Y, 1)$ density, which integrates to 1.

Solution

› Step 1 — marginal.

$$m(Y) = \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-(Y-\mu)^2/2} \cdot 1\,\mathrm{d}\mu = \frac{1}{\sqrt{2\pi}}\sqrt{2\pi} = 1 < \infty.$$

Why? $(Y-\mu)^2 = (\mu - Y)^2$, so the integrand is the $\mathrm{N}(Y, 1)$ density in the variable μ.

› Step 2 — posterior.

$$p(\mu \mid Y) = \frac{f(Y \mid \mu)\pi(\mu)}{m(Y)} = \frac{1}{\sqrt{2\pi}} e^{-(\mu-Y)^2/2},$$

non-negative and integrating to 1: a proper density, namely $\mathrm{N}(Y, 1)$.

Answer

$m(Y) = 1 < \infty$, so $p(\mu \mid Y)$ is proper: $\mu \mid Y \sim \text{Normal}(Y, 1)$.

Takeaway & common mistakes

An improper prior is acceptable only if the posterior is proper — this must be checked model by model (Exercises 16–17 show cases where it fails).

Exercise 2.16 — When an improper prior must fail

Exercise 16 (exact text from the book)

Suppose that the likelihood $f(\mathbf{Y}|\theta) \geq c(\mathbf{Y})$ for all θ. Show that an improper prior on θ will yield an improper posterior.

In plain words

If the likelihood never drops below some positive level, the data cannot "tame" a prior with infinite total mass.

Background you need

Improper prior: $\pi(\theta) \geq 0$ and $\int \pi(\theta)\,\mathrm{d}\theta = \infty$. Proper posterior $\iff$ $m(\mathbf{Y}) = \int f(\mathbf{Y} \mid \theta)\pi(\theta)\,\mathrm{d}\theta < \infty$. If $g \geq h \geq 0$ then $\int g \geq \int h$.

Key idea

Bound the marginal likelihood from below by $c(\mathbf{Y})$ times the (infinite) prior mass.

Solution

Assume $c(\mathbf{Y}) > 0$ (otherwise the bound says nothing). Since $\pi(\theta) \geq 0$, multiplying the inequality $f(\mathbf{Y} \mid \theta) \geq c(\mathbf{Y})$ by $\pi(\theta)$ keeps its direction, and integrating:

$$m(\mathbf{Y}) = \int f(\mathbf{Y} \mid \theta)\pi(\theta)\,\mathrm{d}\theta \;\geq\; \int c(\mathbf{Y})\pi(\theta)\,\mathrm{d}\theta = c(\mathbf{Y}) \int \pi(\theta)\,\mathrm{d}\theta = c(\mathbf{Y}) \times \infty = \infty.$$

So the normalising constant is infinite: $f(\mathbf{Y} \mid \theta)\pi(\theta)$ cannot be scaled into a density, i.e. the posterior is improper.

Answer

$m(\mathbf{Y}) \geq c(\mathbf{Y}) \int \pi(\theta)\,\mathrm{d}\theta = \infty$ whenever $c(\mathbf{Y}) > 0$, so the posterior is improper.

Takeaway & common mistakes

For a proper posterior under an improper prior, the likelihood must *decay* in every direction where the prior has infinite mass.

Exercise 2.17 — Mixture models and improper priors

Exercise 17 (exact text from the book)

Consider a sample of size n from the mixture of two normal densities given by $f(y|\theta) = 0.5\phi(y) + 0.5\phi(y-\theta)$ where $\phi(z)$ denotes the standard normal density function. Show that any improper prior for θ leads to an improper posterior.

In plain words

Half the data come from $\mathrm{N}(0,1)$ and half from $\mathrm{N}(\theta,1)$. Show that no improper prior on θ can work.

Background you need

Exercise 16: a likelihood bounded below by a positive constant $c(\mathbf{y})$ makes every improper prior give an improper posterior. $\phi(z) > 0$ for every z.

Key idea

Find a lower bound for the likelihood that does not involve θ.

Solution

› Step 1 — one observation.

Since $\phi(y_i - \theta) \geq 0$,

$$f(y_i \mid \theta) = 0.5\phi(y_i) + 0.5\phi(y_i - \theta) \;\geq\; 0.5\phi(y_i) \quad \text{for every } \theta.$$

› Step 2 — the whole sample.

The observations are independent, so the likelihood is a product of positive numbers each bounded below:

$$f(\mathbf{y} \mid \theta) = \prod_{i=1}^{n} f(y_i \mid \theta) \;\geq\; \prod_{i=1}^{n} 0.5\phi(y_i) =: c(\mathbf{y}) > 0.$$

› Step 3 — conclude.

By Exercise 16, $m(\mathbf{y}) \geq c(\mathbf{y}) \int \pi(\theta)\,\mathrm{d}\theta = \infty$.

Answer

$f(\mathbf{y} \mid \theta) \geq \prod_i 0.5\phi(y_i) > 0$ for all θ, so by Exercise 16 every improper prior gives an improper posterior.

Takeaway & common mistakes

Intuition: as $\theta \to \pm\infty$ the likelihood does not vanish — every observation could have come from the first component, which says nothing about θ. Mixture models need proper priors (see Chapter 3, Exercise 15).

Exercise 2.18 — An evidence-based prior for clinical trials

Exercise 18 (exact text from the book)

Say a randomized clinical trial is conducted to compare the difference in the mean response between two treatment groups. The effect size b is estimated as the difference between the sample means of the two groups, and b has standard error s, and thus the z-score is $z = b/s$. The true difference between treatment group means is β, and the signal-to-noise ratio is $\delta = \beta/s$. We fit the model $z|\delta \sim \text{Normal}(\delta, 1)$ and δ has mixture prior in (2.33).

(a) Plot the posterior distribution of δ if we observe $z = 2$.

(b) Compute the posterior probability that $\delta > 0$ given $z = 2$.

(c) Compute the posterior probability that $\delta > 0$ for value of $z \in \{0.0, 0.1, \ldots, 5.0\}$ and plot the probabilities by z.

(d) Repeat these analyses using prior $\delta \sim \text{Normal}(0, 100^2)$, and comment on the effect of the prior.

In plain words

Using a prior learned from 20,000 real trials, update the true signal-to-noise ratio δ after seeing z, and compare with a nearly flat prior.

Background you need

Prior (2.33) is a mixture of four zero-mean normals:

$$\pi(\delta) = \sum_{k=1}^{4} w_k \, \phi(\delta; 0, \tau_k^2), \qquad w = (0.32,\ 0.31,\ 0.30,\ 0.07), \quad \tau = (1.17,\ 1.74,\ 2.38,\ 5.73).$$

For one component: $z \mid \delta \sim \text{N}(\delta, 1)$, $\delta \sim \text{N}(0, \tau^2)$ give $\delta \mid z \sim \text{N}\big(\frac{\tau^2}{1+\tau^2} z,\ \frac{\tau^2}{1+\tau^2}\big)$, and marginally $z \sim \text{N}(0, 1 + \tau^2)$.

Key idea

Mixture update (primer, point 5): the posterior is a mixture of the four component posteriors, with weights $w'_k \propto w_k \, \phi(z; 0, 1 + \tau_k^2)$.

Solution

(a) Step 1 — per-component posteriors for $z = 2$.

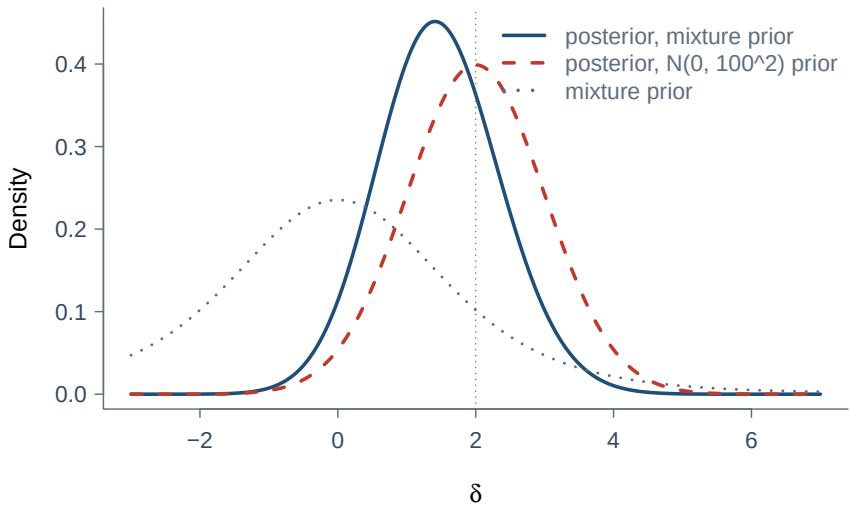
Shrinkage factor $s_k = \frac{\tau_k^2}{1+\tau_k^2}$; component mean $s_k z$; component variance s_k. For example $k = 1$: $s_1 = \frac{1.3689}{2.3689} = 0.578$, mean $0.578 \times 2 = 1.156$.

(a) Step 2 — updated weights.

$w'_k \propto w_k \times$ (density of $z = 2$ under $\text{N}(0, 1 + \tau_k^2)$); normalise to sum to 1.

Component	1	2	3	4
prior weight w_k	0.32	0.31	0.30	0.07
posterior weight w'_k	0.318	0.335	0.307	0.040
shrinkage s_k	0.578	0.752	0.850	0.970
component mean $s_k z$	1.156	1.503	1.700	1.941

Posterior mean $\sum_k w'_k \, s_k z = 1.47$.

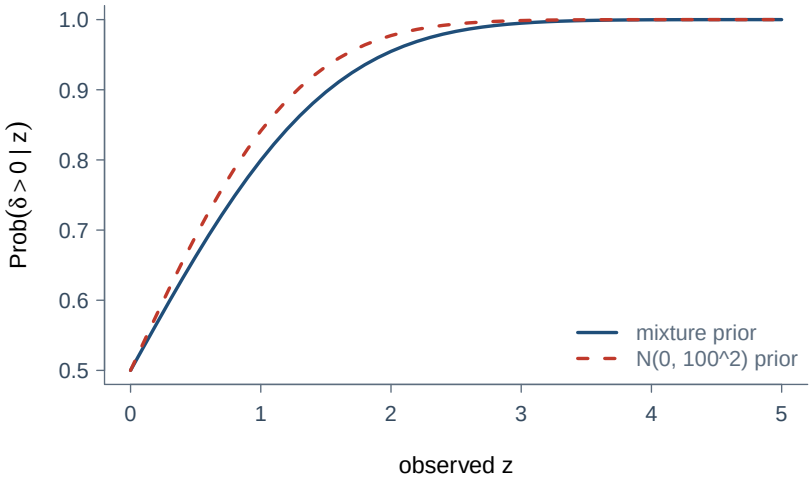


(b)

$\text{Prob}(\delta > 0 \mid z = 2) = \sum_k w'_k \, \Phi\left(\frac{s_k z}{\sqrt{s_k}}\right) = \mathbf{0.955}$.

Why? For a normal component with mean m and SD s, $\text{Prob}(\delta > 0) = \Phi(m/s)$; then average over components with the posterior weights.

(c) Across z.



(d) Vague prior $\text{N}(0, 100^2)$.

Shrinkage $\frac{10^4}{1+10^4} \approx 1$, so $\delta \mid z \approx \text{N}(z, 1)$: posterior mean 2.00, $\text{Prob}(\delta > 0 \mid z = 2) = \Phi(2) = 0.977$ (dashed curves above).

z	1	1.96	2	3	4
mixture prior	0.800	0.951	0.955	0.995	0.9997
$\text{N}(0, 100^2)$ prior	0.841	0.975	0.977	0.999	1.000

Sanity check

At $z = 0$ both curves give exactly 0.5 (symmetric prior, no evidence either way). ✓

Answer

(a) A four-component normal mixture centred at 1.47. (b) $\text{Prob}(\delta > 0 \mid z = 2) = 0.955$. (c) Rises from 0.5 at $z = 0$ to ≈ 1 by $z \approx 4$. (d) The vague prior gives $\delta \mid z \approx \text{N}(z, 1)$, $\text{Prob}(\delta > 0 \mid z = 2) = 0.977$ — no shrinkage; the evidence-based prior shrinks the estimate by $\approx 25\%$ and is less confident.

Takeaway & common mistakes

The evidence-based prior corrects the "winner's curse": a trial that just reaches significance usually overestimates the true effect. Note that $z = 1.96$ ("$p = 0.05$") gives only $\text{Prob}(\delta > 0) = 0.95$, not 0.975.

Exercise 2.19 — Jeffreys prior for the Poisson

Exercise 19 (exact text from the book)

Say $Y|\lambda \sim \text{Poisson}(\lambda)$.

(a) Derive and plot the Jeffreys prior for λ.

(b) Is this prior proper?

(c) Derive the posterior, and give conditions on Y to ensure it is proper.

In plain words

Compute the Fisher information of the Poisson, turn it into the Jeffreys prior, and check whether prior and posterior are proper.

Background you need

Jeffreys prior: $\pi(\lambda) \propto \sqrt{I(\lambda)}$, $I(\lambda) = -\text{E}\left[\frac{\partial^2}{\partial\lambda^2}\log f(Y \mid \lambda)\right]$. For a Poisson, $\text{E}Y = \lambda$. A Gamma(a, b) is a proper distribution iff $a > 0$ and $b > 0$.

Key idea

Differentiate the log-likelihood twice, take the expectation over Y, then take a square root.

Solution

(a) Step 1 — log-likelihood.

$\log f(Y \mid \lambda) = Y \log\lambda - \lambda - \log Y!$.

(a) Step 2 — derivatives.

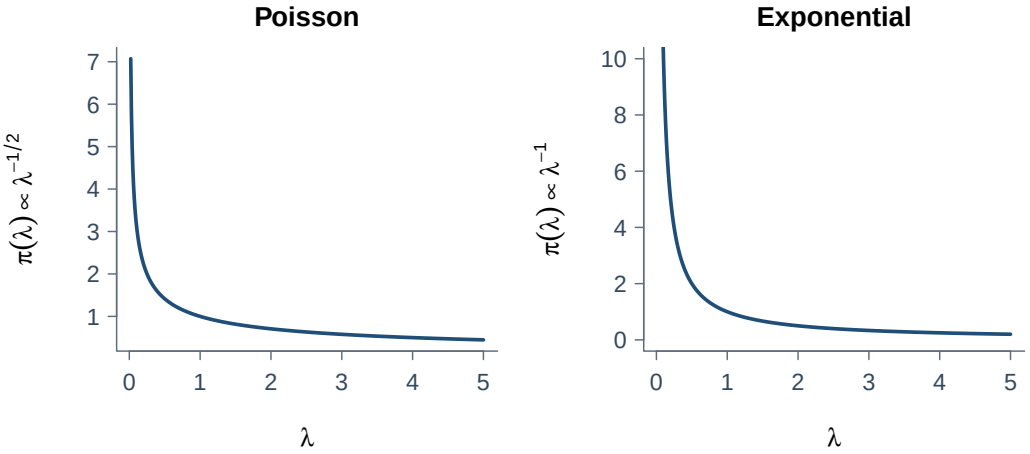
$\frac{\partial}{\partial\lambda}\log f = \frac{Y}{\lambda} - 1$; $\frac{\partial^2}{\partial\lambda^2}\log f = -\frac{Y}{\lambda^2}$.

(a) Step 3 — Fisher information.

$I(\lambda) = -\text{E}\left(-\frac{Y}{\lambda^2}\right) = \frac{\text{E}Y}{\lambda^2} = \frac{\lambda}{\lambda^2} = \frac{1}{\lambda}$.

(a) Step 4 — Jeffreys prior.

$\pi(\lambda) \propto \sqrt{1/\lambda} = \lambda^{-1/2}$, $\lambda > 0$.



(b) Proper?

$\int_0^\infty \lambda^{-1/2}\,\mathrm{d}\lambda = \big[2\sqrt{\lambda}\big]_0^\infty = \infty$. **Improper** (the upper tail is the problem).

(c) Posterior.

$$p(\lambda \mid Y) \propto \lambda^Y e^{-\lambda} \cdot \lambda^{-1/2} = \lambda^{(Y+\frac{1}{2})-1} e^{-\lambda} \;\Rightarrow\; \lambda \mid Y \sim \mathrm{Gamma}\big(Y + \tfrac{1}{2},\ 1\big).$$

Shape $Y + \frac{1}{2} > 0$ and rate $1 > 0$ for every $Y \in \{0, 1, 2, \dots\}$, so the posterior is always proper.

Answer

(a) $\pi(\lambda) \propto \lambda^{-1/2}$. (b) Improper. (c) $\lambda \mid Y \sim \mathrm{Gamma}(Y + \frac{1}{2}, 1)$, proper for **every** $Y \geq 0$, including $Y = 0$.

Takeaway & common mistakes

The Jeffreys prior is the improper "$\mathrm{Gamma}(\frac{1}{2}, 0)$"; it is the default prior used in Exercise 21.

Exercise 2.20 — Jeffreys prior for the exponential rate

Exercise 20 (exact text from the book)

Say $Y|\lambda \sim \text{Gamma}(1, \lambda)$.

(a) Derive and plot the Jeffreys prior for λ.

(b) Is this prior proper?

(c) Derive the posterior, and give conditions on Y to ensure it is proper.

In plain words

Same as Exercise 19, for an exponential waiting time with rate λ.

Background you need

$\text{Gamma}(1, \lambda)$ is the exponential with rate λ: $f(y \mid \lambda) = \lambda e^{-\lambda y}$, $y > 0$.

Key idea

Same recipe as Exercise 19. Here the second derivative does not even involve Y, so no expectation is needed.

Solution

› (a) Step 1 — log-likelihood.

$\log f = \log \lambda - \lambda Y$.

› (a) Step 2 — derivatives.

$\frac{\partial}{\partial \lambda} \log f = \frac{1}{\lambda} - Y$; $\frac{\partial^2}{\partial \lambda^2} \log f = -\frac{1}{\lambda^2}$.

› (a) Step 3 — Jeffreys prior.

$I(\lambda) = \frac{1}{\lambda^2} \Rightarrow \pi(\lambda) \propto \frac{1}{\lambda}$ (right panel of the figure in Exercise 19).

› (b) Proper?

$\int_0^\infty \lambda^{-1} \, \mathrm{d}\lambda = \big[\log \lambda\big]_0^\infty$ diverges at *both* ends. **Improper.**

› (c) Posterior.

$$p(\lambda \mid Y) \propto \lambda e^{-\lambda Y} \cdot \lambda^{-1} = e^{-\lambda Y} \;\Rightarrow\; \lambda \mid Y \sim \text{Gamma}(1,\ Y) = \text{Exponential}(Y).$$

Proper iff the rate $Y > 0$. Since Y is a positive continuous variable, $Y > 0$ with probability 1. (If $Y = 0$, $p \propto 1$ would be improper.)

Answer

(a) $\pi(\lambda) \propto 1/\lambda$. (b) Improper. (c) $\lambda \mid Y \sim \text{Gamma}(1, Y)$, proper iff $Y > 0$.

Takeaway & common mistakes

$1/\lambda$ is the classic scale-invariant prior (flat on $\log \lambda$), the Jeffreys prior for any rate or scale parameter.

Exercise 2.21 — A default analysis of one Poisson count

Exercise 21 (exact text from the book)

Assume the model $Y|\lambda \sim \text{Poisson}(\lambda)$, and we observe $Y = 10$. Conduct a Bayesian analysis of these data with default prior distribution.

In plain words

With a single count of 10 and no prior information, use an objective prior and summarise the posterior.

Background you need

From Exercise 19: Jeffreys prior $\pi(\lambda) \propto \lambda^{-1/2}$ gives $\lambda \mid Y \sim \text{Gamma}(Y + \frac{1}{2}, 1)$.

Key idea

Use the Jeffreys prior (the standard default for a Poisson rate), and compare with other common defaults.

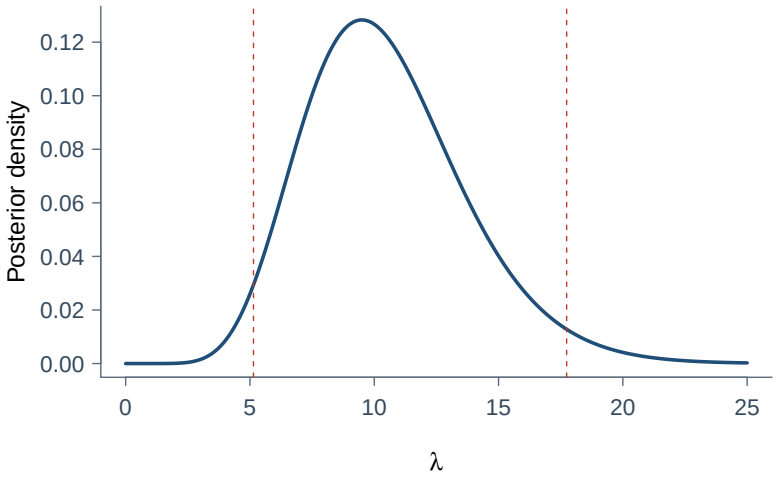
Solution

› Step 1 — posterior.

$\lambda \mid Y = 10 \sim \text{Gamma}(10.5,\ 1)$.

› Step 2 — summaries.

Mean $10.5/1 = 10.5$; SD $\sqrt{10.5}/1 = 3.24$; 95% interval $= (5.14,\ 17.74)$ (`qgamma(c(.025, .975), 10.5, 1)`).



› Step 3 — other default priors.

Prior	Posterior	Mean	95% interval
Jeffreys $\propto \lambda^{-1/2}$	$\text{Gamma}(10.5, 1)$	10.50	(5.14, 17.74)
Flat $\propto 1$	$\text{Gamma}(11, 1)$	11.00	(5.49, 18.39)
$\text{Gamma}(0.01, 0.01)$	$\text{Gamma}(10.01, 1.01)$	9.91	(4.75, 16.93)

Answer

Jeffreys prior $\Rightarrow \lambda \mid Y \sim \text{Gamma}(10.5, 1)$: posterior mean 10.5, SD 3.2, 95% credible interval $(5.1,\ 17.7)$.

Takeaway & common mistakes

With a single observation, "default" priors still differ by about half a unit in the mean. Default does not mean irrelevant — report the choice.

Exercise 2.22 — HOV lane survey: Jeffreys vs. empirical Bayes

Exercise 22 (exact text from the book)

The data in the table below are the result of a survey of commuters in 10 counties likely to be affected by a proposed addition of a high occupancy vehicle (HOV) lane.

County	Approve	Disapprove	County	Approve	Disapprove
1	12	50	6	15	8
2	90	150	7	67	56
3	80	63	8	22	19
4	5	10	9	56	63
5	63	63	10	33	19

(a) Analyze the data in each county separately using the Jeffreys prior distribution, and report the posterior 95% credible set for each county.

(b) Let $\hat{p}_i$ be the sample proportion of commuters in county i that approve of the HOV lane (e.g., $\hat{p}_1 = 12/(12+50) = 0.194$). Select a and b so that the mean and variance of the $\text{Beta}(a, b)$ distribution match the mean and variance of the sample proportions $\hat{p}_1, \ldots, \hat{p}_{10}$.

(c) Conduct an empirical Bayesian analysis by computing the 95% posterior credible sets that results from analyzing each county separately using the $\text{Beta}(a, b)$ prior you computed in (b).

(d) How do the results from (a) and (c) differ? What are the advantages and disadvantages of these two analyses?

In plain words

Estimate approval in each county two ways — alone with an objective prior, and with a prior estimated from all counties together — and compare.

Background you need

- Binomial Jeffreys prior: $\text{Beta}(\frac{1}{2}, \frac{1}{2})$.
- Beta moment matching (Exercise 13): $a + b = \frac{m(1-m)}{v} - 1$, $a = m(a+b)$, $b = (1-m)(a+b)$.

Key idea

Empirical Bayes = estimate the prior from the spread of the county proportions, then use it in every county. Small counties get pulled toward the overall average (**shrinkage**).

Solution

(a) Jeffreys analysis.

Each county separately: $\theta_i \mid Y_i \sim \text{Beta}(Y_i + \frac{1}{2},\ n_i - Y_i + \frac{1}{2})$. E.g. county 1: $\text{Beta}(12.5,\ 50.5)$. The 95% intervals are in the left half of the table below.

(b) The empirical prior.

Mean of the ten $\hat{p}_i$: $m = 0.480$; variance: $v = 0.02026$. Then

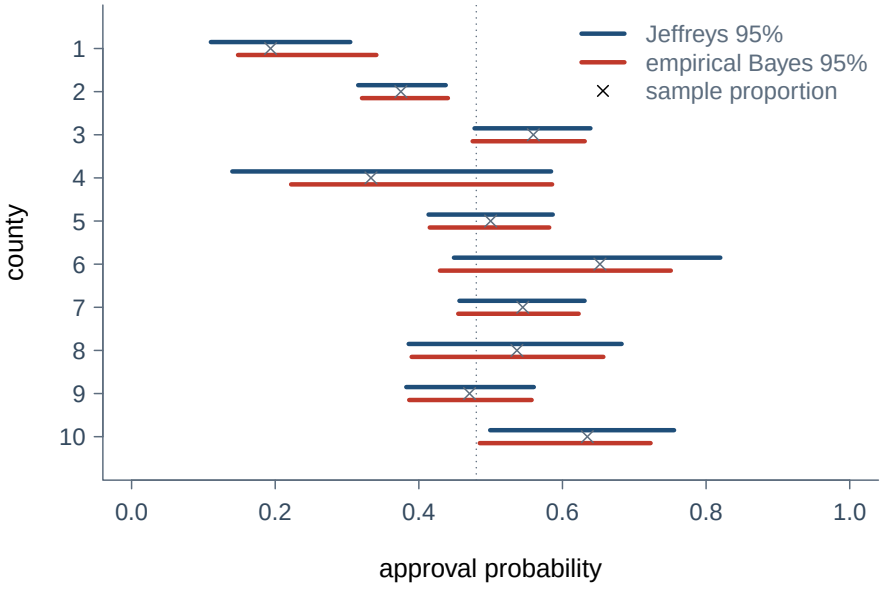
$$a + b = \frac{0.480 \times 0.520}{0.02026} - 1 = 11.32, \qquad a = 0.480 \times 11.32 = 5.43, \qquad b = 0.520 \times 11.32 = 5.89.$$

So the prior is $\text{Beta}(5.43,\ 5.89)$: mean 0.48, worth about 11 respondents.

(c) Empirical Bayes analysis.

Each county with the $\text{Beta}(5.43, 5.89)$ prior: $\theta_i \mid Y_i \sim \text{Beta}(5.43 + Y_i,\ 5.89 + n_i - Y_i)$ (right half of the table).

			(a) Jeffreys		(c) Empirical Bayes	
County	n_i	$\hat{p}_i$	mean	95% interval	mean	95% interval
1	62	0.194	0.198	(0.110, 0.305)	0.238	(0.148, 0.341)
2	240	0.375	0.376	(0.316, 0.437)	0.380	(0.321, 0.440)
3	143	0.559	0.559	(0.478, 0.639)	0.554	(0.475, 0.631)
4	15	0.333	0.344	(0.140, 0.584)	0.396	(0.222, 0.586)
5	126	0.500	0.500	(0.414, 0.586)	0.498	(0.415, 0.582)
6	23	0.652	0.646	(0.449, 0.820)	0.595	(0.429, 0.751)
7	123	0.545	0.544	(0.457, 0.631)	0.539	(0.455, 0.623)
8	41	0.537	0.536	(0.386, 0.682)	0.524	(0.390, 0.657)
9	119	0.471	0.471	(0.383, 0.560)	0.471	(0.387, 0.557)
10	52	0.635	0.632	(0.499, 0.755)	0.607	(0.485, 0.723)



Why? EB posterior mean $= \frac{5.43+Y_i}{11.32+n_i}$: a weighted average of the prior mean 0.48 (weight $\frac{11.32}{11.32+n_i}$) and $\hat{p}_i$. Small $n_i \Rightarrow$ more weight on 0.48.

(d) Comparison.

- **Shrinkage**: EB pulls estimates toward 0.48, most for small counties (county 4, $n = 15$: $0.344 \to 0.396$; county 6, $n = 23$: $0.646 \to 0.595$), hardly at all for large ones (county 2, $n = 240$).
- **Precision**: EB intervals are narrower (average width 0.216 vs. 0.237).
- **Jeffreys** — pros: objective, each county stands alone, no assumption that counties are similar. Cons: ignores information from other counties; wide intervals for small counties.
- **Empirical Bayes** — pros: borrows strength, better estimates on average for small counties. Cons: uses the data twice and treats a, b as known, so its intervals are somewhat too narrow; assumes counties are exchangeable.

Answer

(a) Jeffreys intervals: see table. (b) $a = 5.43$, $b = 5.89$. (c) EB intervals: see table.
(d) EB shrinks small counties toward 0.48 and narrows intervals, at the cost of ignoring uncertainty in (a, b). The fully Bayesian fix is a *hierarchical model* with priors on a, b (Section 2.2 / Chapter 7).

Takeaway & common mistakes

Empirical Bayes sits between "each county alone" and "all counties identical"; the data decide the amount of shrinkage (here weak: about 11 pseudo-respondents).